



Atag Celcius induction cooker

Phase 1 | Description, Evaluation and
Design Brief

Team 12

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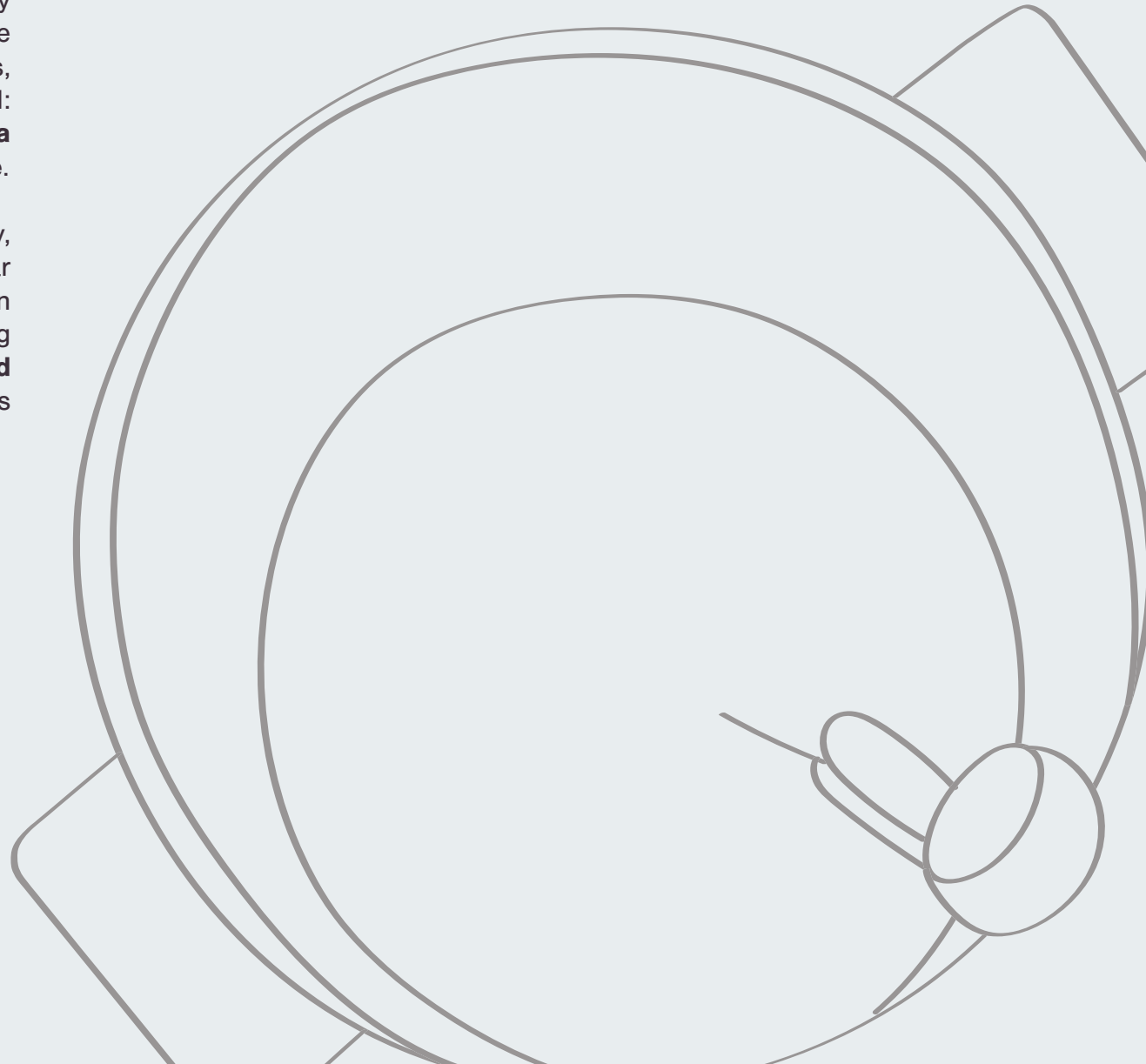
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Abstract

The Atag Celsius induction hob represents the prime of the brand's **high-end kitchen appliances**, designed to make cooking simpler and more precise with smart temperature control. This project aimed to assess its practicality through **design assessment methods** including cognitive walkthroughs, questionnaires, user tests, market analysis, and reviews. Identified challenges in daily use led to a goal: **redesign the hob for enhanced usability, targeting a dependable and convenient cooking experience at home.**

The report encapsulates our initial research and methodology, laying the groundwork for informed design decisions. Clear design objectives and an interaction vision have been established to steer the project. A structured plan for upcoming stages aligns with our overarching goals, ensuring a **focused approach to improving the Atag Celsius** induction hob's user experience.



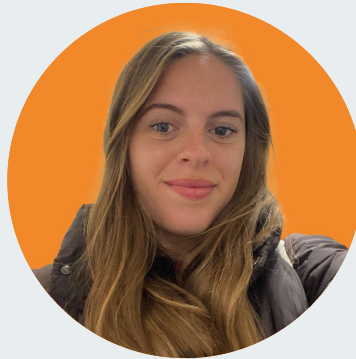
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1 ■ Product description

This chapter describes the Atag Celsius induction cooker. Different analysis methods are used to describe the function and details of the product. The subchapters should provide a good understanding of the product and the company.

Product introduction
Company
Product build up
Intended operation
Users
Context of use



Product introduction



The Atag Celsius induction hob represents an advancement in kitchen technology, tailored for culinary professionals seeking accurate temperature management and a range of cooking techniques. Featuring advanced temperature regulation technology, this induction cooker provides precise control over cooking temperatures. It monitors food temperature in real-time via an included probe or a specific pan, ensuring optimal cooking outcomes.



The inclusion of Celsius° Cooking technology enhances its adaptability, offering three cooking modes that enable the customization of dishes according to various ingredients and recipe requirements. Its user-friendly interface combines buttons and a touch screen, simplifying operation. Further personalization is made possible through integration with a dedicated app, allowing users to tailor and save their cooking programs based on personal taste and needs, thus achieving a truly customized cooking experience.

The Company: ATAG

ATAG is a premium home kitchen company based in Achterhoek in the Netherlands. The company started in 1948 with gas stoves to meet post-war needs. Nowadays, the company has grown strongly in great quality and many innovations. Their motto is as followed: '**Discover the chef within yourself**'. ATAG wants to encourage people to experiment with cooking. In 1991, the first induction cooking plate from ATAG was introduced. Induction cooking was especially popular in Germany and Belgium. These two countries became the biggest clients of ATAG. After improving marketing, sales in the Netherlands started to increase. New unique innovations led to the Celsius cooking plate: cooking with temperatures.

Customers

ATAG's high-quality products come with a price. ATAG's customers have a bigger budget than average. Although ATAG is marketing with three user-groups, the social chef, semi pro and pleasure seeker, their actual customer base is more diverse. They keep in mind that anyone can cook, not only the purchaser but also, for example, the children and visitors.

Brand values

ATAG has multiple brand values that are important when developing and improving their products. Figure 1 shows their brand values, which are arrived from ATAG's brand videos and website.



Figure 1: Brand Values of Atag.

Product portfolio

85

ATAG has an extensive product portfolio. They sell equipment for the whole kitchen. Their collection is as followed:

- Gas cooking plates
- Induction cooking plates
- Extractor hoods (on the wall and in the counter)
- Dishwasher
- Ovens
- Microwaves
- Fridges and freezers
- Build-in coffee machines
- Wine coolers
- Small accessoires

As mentioned in a client meeting ATAG has three classes of products. Good, better and best. These classes differ from prices and features. For the induction plate, three classes are shown on the left.

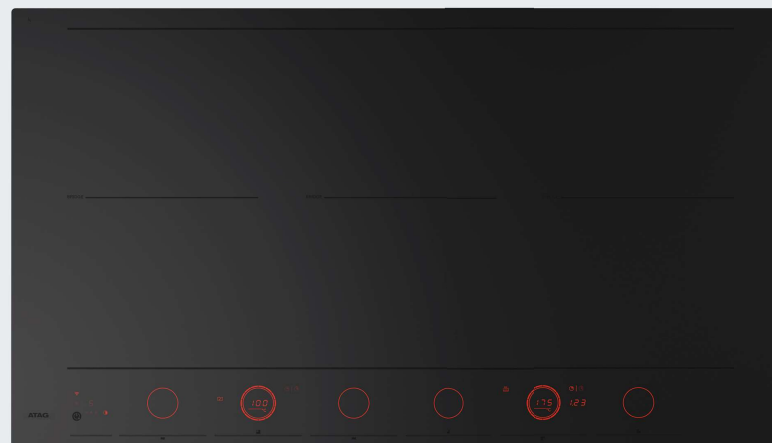


Good
Induction plate MAGNA
Price: €999,-

- Direct slide control
- Timer per hob
- Pan detection

Better
Induction plate MAGNA
Price: €2.099,-

- Touch control per hob
- Automatically bridging two hobs
- Temperature programs
- Timer per hob
- Pan detection



Best
Induction with celsius cooking
Price: €2.799,-

- Iris slide control per hob
- Automatically bridging two hobs
- Plus and celsius settings
- Timer and stopwatch per hob
- Pan detection
- Connected to the ConnectLife app

Brand positioning

Before starting with the analysis of the ATAG celsius cooking plate, it is good to find out what sets ATAG apart from other companies.

In Figure 2 on the left there can be seen that ATAG is a very high end brand, they strive for quality within their products and innovation. On the scale from formal to playful ATAG scores a bit more to the formal side. Their products and website all look quite more formal. Although ATAG's marketing is more playful. They market their products in context of a cosy home feeling. Where ATAG is different from other companies is their goal to encourage people to explore cooking and improving their skills at home.

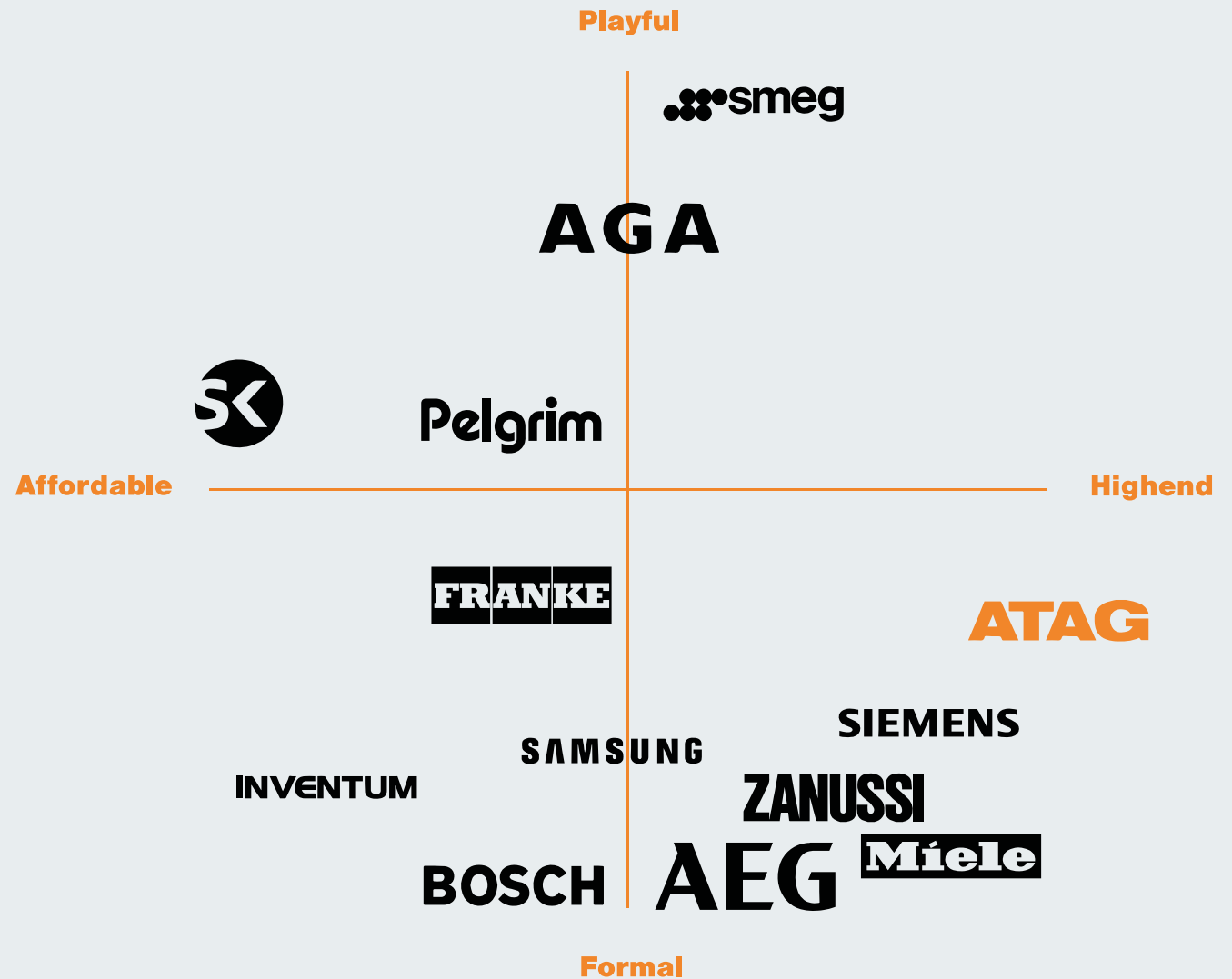


Figure 2: C box of Kitchen Brands.

Product build up

To effectively redesign the Atag Celsius cooking hob, we need a clear understanding of its construction. To achieve this, we have created a breakdown of the product's various components. In the following pages, we have distinguished the different interaction areas, including the hob touch interface, digital interface, and Celsius cooking accessories. We have also explored the technical composition of the product to gain a deeper understanding of how it works and functions.



Main interface

The cooking hob's main interface is the first thing the user will see upon starting the interaction (see Figure 3). The hob's functions are only visible when turned on, which is designed to minimize confusion. As a result, the active or available functions are no longer printed on the glass but will light up when in use or available to use. This creates a smaller area for the user's first interaction, reducing unnecessary complexity.

- 1 Cookingzone D21cm
- 2 Bridge cookingzone 18x22cm
- 3 Cookingzone D18 cm
- 4 Control Panel (only visible when hob is switched on)
- 5 On/Off Switch

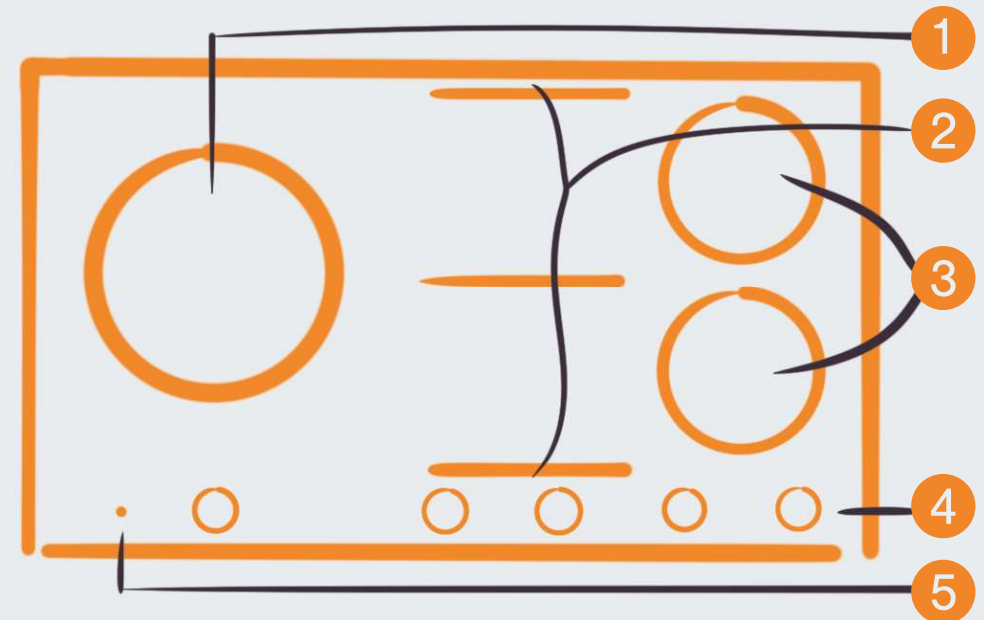


Figure 3: Overview of the main interface.

Digital Interface

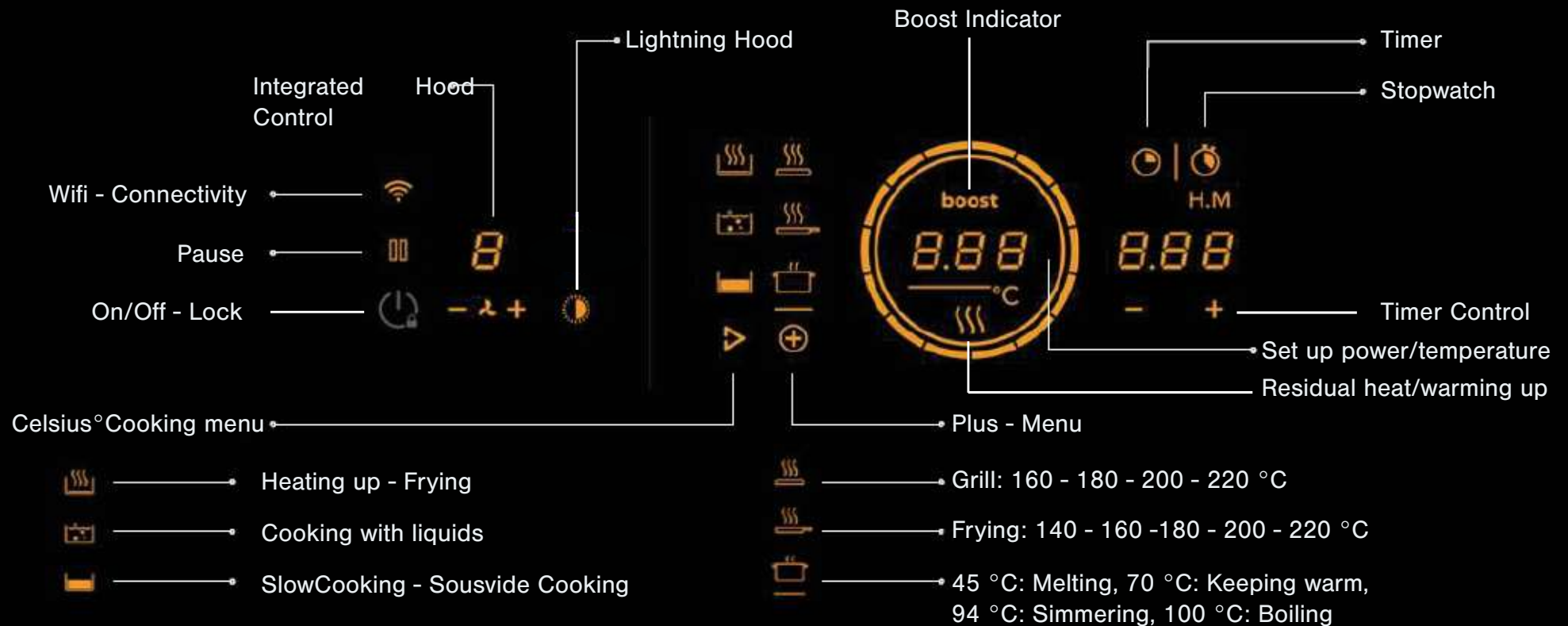


Figure 4: Digital Interface

The hob's digital interface (see Figure 4) is what connects the user to the cooking experience. The image above displays the main controls, while personal settings and error codes are relegated to a subsequent page, as they are not part of the daily interaction. The interface allows users to choose their cooking settings, set a timer or stopwatch, connect to wifi, pause cooking, and lock the screen

to keep children safe or for easy cleaning. If the user has a model with a connected fan, they can adjust and connect it through this interface as well, though we will not be exploring this function in this research. Additionally, the screen displays valuable information such as current temperature, warming state, residual heat, active timer, or any system errors.

Accessories

Atag has introduced accessories for Celsius Cooking with this hob series. The accessories, including the probe and Celsius Cooking Skillet, connect to the hob via Bluetooth to ensure the food's exact temperature. Apart from these advanced gadgets, the hob also comes with a small ruler to allow the user to determine the 'mass' of their roast.

Skillet

The Atag Celcius Cooking skillet is a 3-ply stainless steel bottom and is equipped with a smart sensor that measures the temperature of the bottom of the pan accurately. To activate the celcius cooking mode you should double tap the Celcius Cooking Icon on the end of the handle. This is also the part you should screw open to insert one AAA battery.



Probe

The ATAG Probe is a smart (core) thermometer that communicates with the hob. You can attach the stainless steel fastening clip to any pan and measure its temperature. Alternatively, you can insert the probe directly into the product you're cooking to measure its exact core temperature. The probe comes with two 3V batteries, of which only one is needed, and you can insert them by opening the back. To activate the probe, simply tap on the Celsius Cooking Icon on the front.

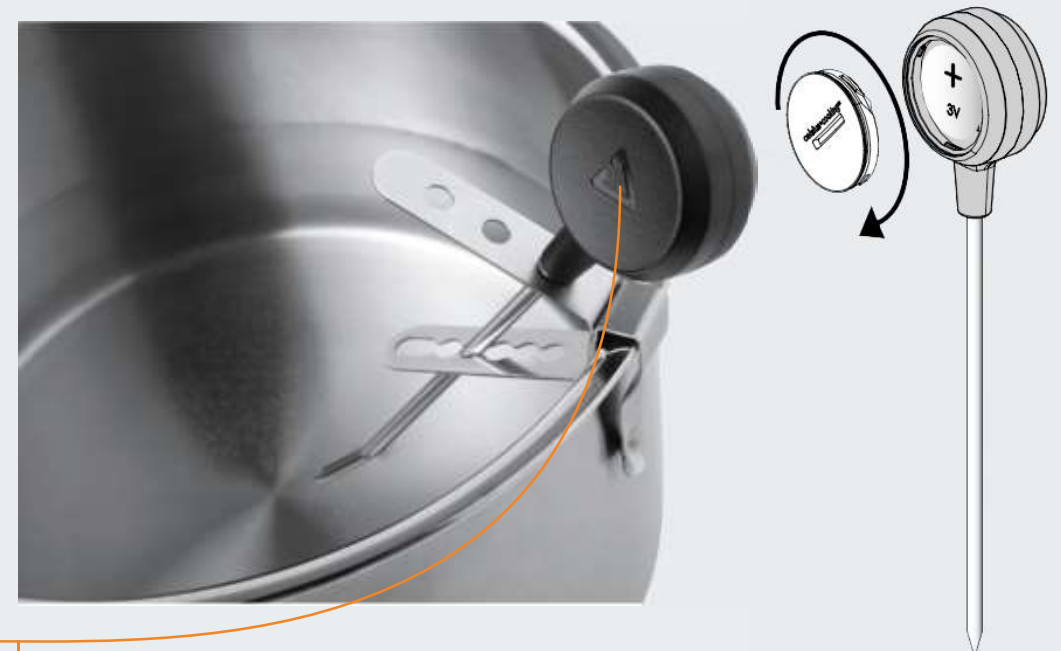


Figure 5: Accessoires of Celcius Cooking

CelciusCooking Icon

Technical Buildup

In order to gain a deeper comprehension of the product, we delved into the technical aspects that constituted its construction which is depicted in Figure 6.

1. Glass/ Ceramic Plate
2. Sealing Tape
3. Temperature Sensor
4. Induction coil D180,200 mm
5. Induction coil D180.460 mm
6. Coil RV
7. Induction coil D260.300 mm
8. Coil LV
9. Other fasting or insulating material
10. Cable Gen-Gen
11. Genarator E.G.O.
12. Generator G5
13. Power supply
14. Cable Wiringharness TCM-TCS
15. Lid Terminal Block
16. Wifi Module
17. Cable Wiringharness WIFI
18. Pin Module
19. HSC-Module
20. Cable Wiringharness LIN
21. Board Operating Master
22. Board Operating Slave
23. Cable Wiringharness IRIS

Under the Surface

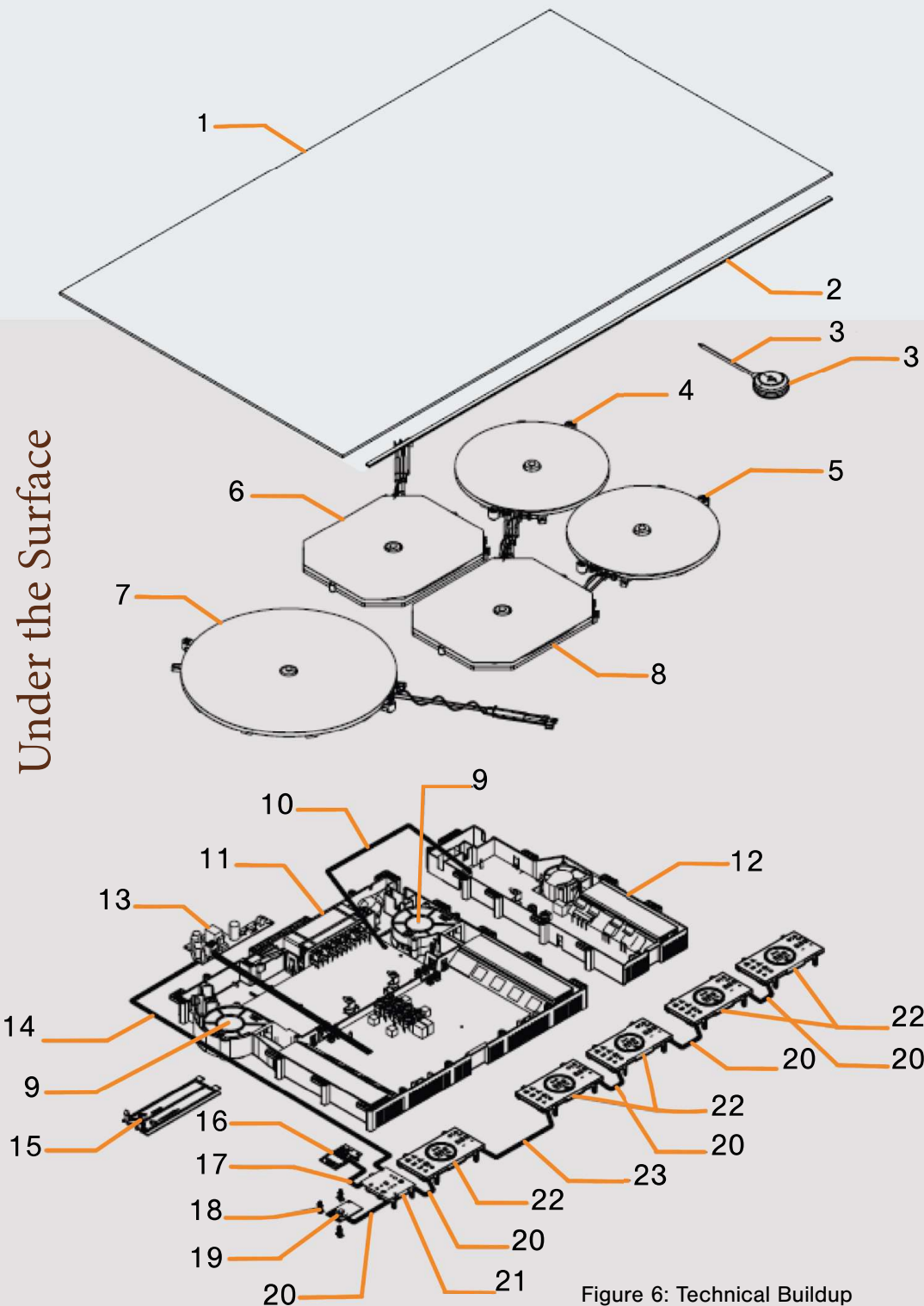


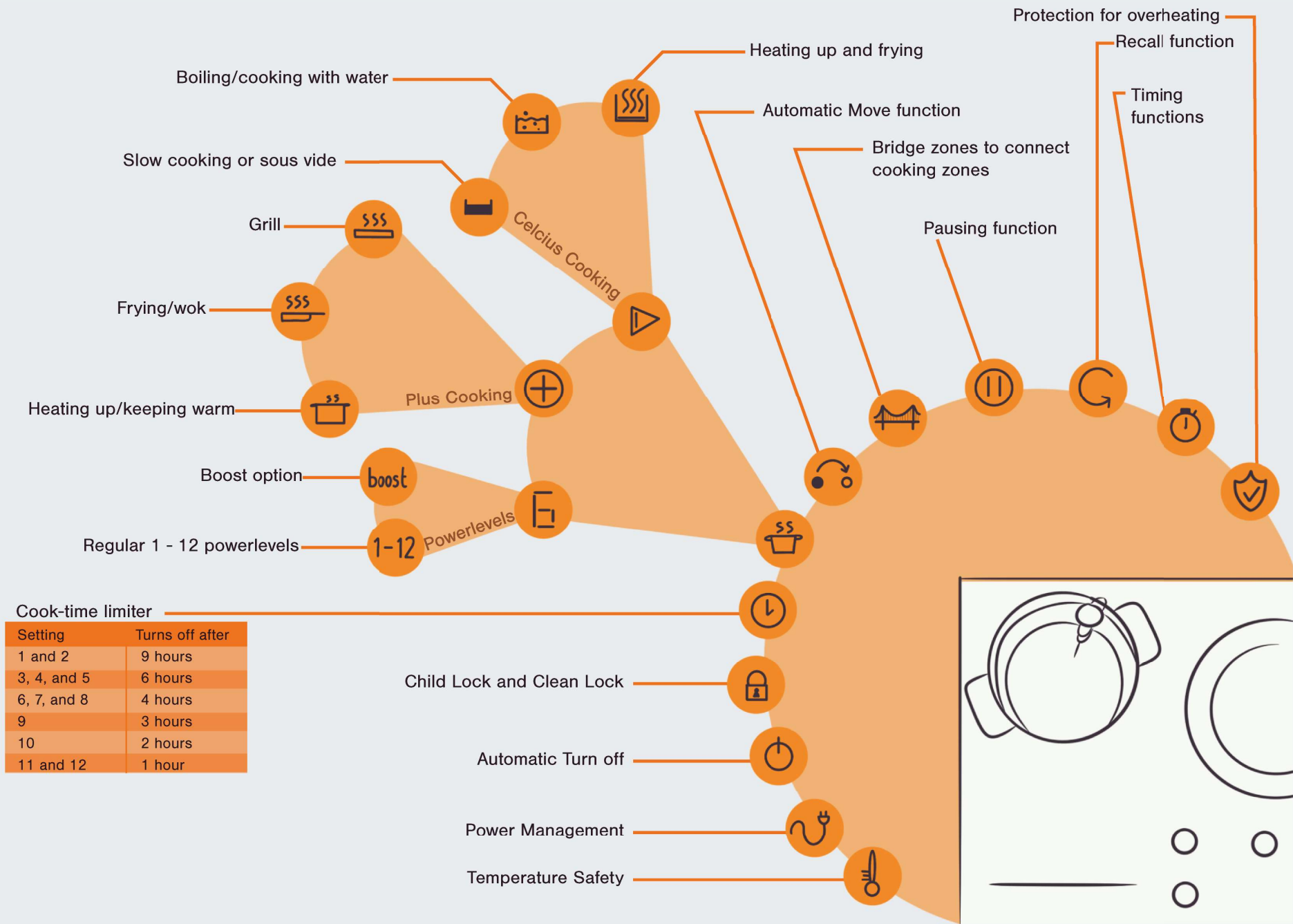
Figure 6: Technical Buildup

The App

Atag collaborated with Hestan to develop the CelciusCooking line and an app that allows users to connect recipes and monitor the hob from a distance, but users couldn't control the hob through the app due to safety reasons. The collaboration ended, and Atag is now focused on developing the ConnectLife app that will enable users to manage and monitor their smart homes easily from anywhere, anytime. The app will allow users to send recipes to the hob and receive guidance on how to prepare them by automatically adjusting the hob's settings. However, the development of the app is still ongoing. Therefore, it is excluded from our scope for the rest of the analysis of the product.



Overview of functions



Intended Operation

Atag has developed the Celcius Cooking induction cooktop with a wide range of functions that can be complex for users. A detailed task flow analysis has been conducted to address this issue, and the whole task flow with interconnection between the areas can be found in Appendix A; below is a simplified version depicted. During our conversation with Atag, they suggested that the hob is designed with a distinction between primary functions, such as cooking on power levels, and secondary functions, such as PLUS cooking and Celcius Cooking. Therefore, based on this distinction, we have separated the user flows into building blocks a simplified version is depicted in Figure 7.

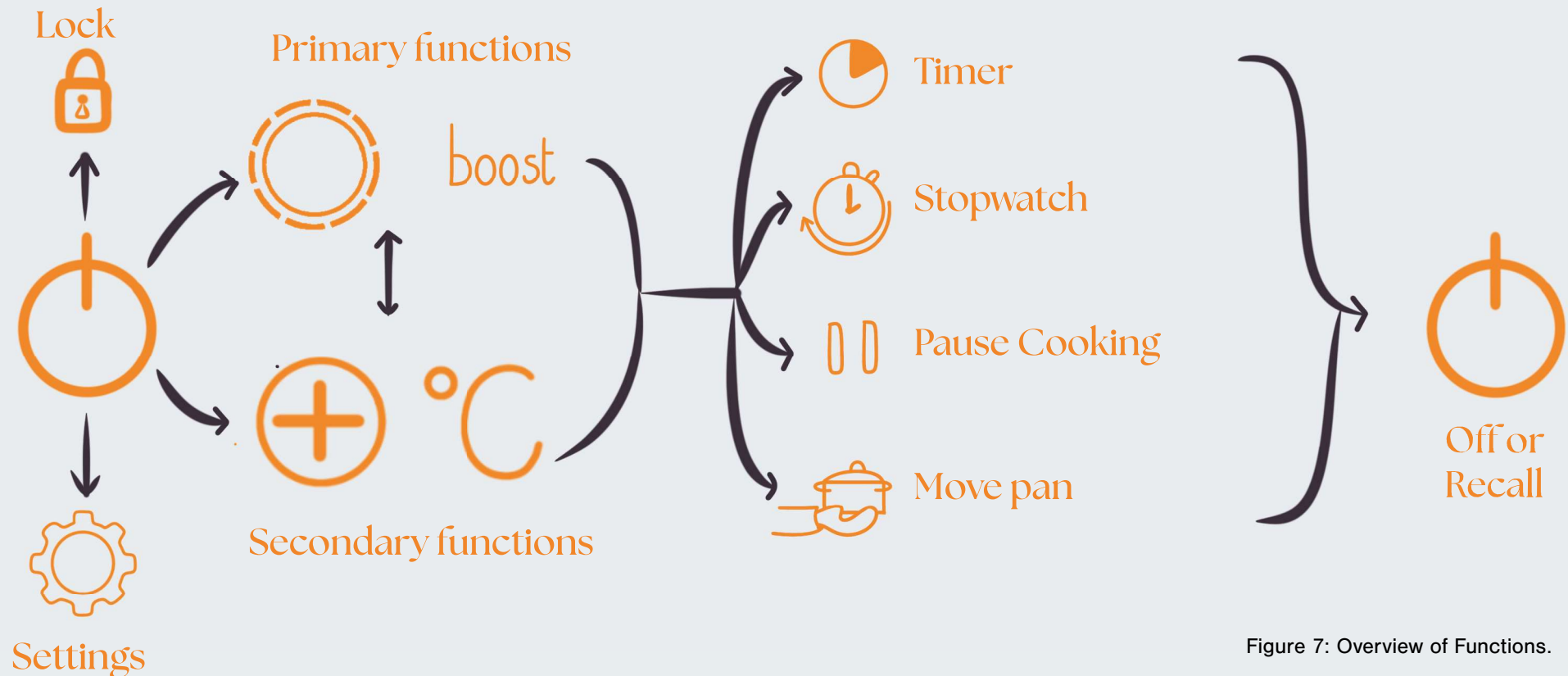
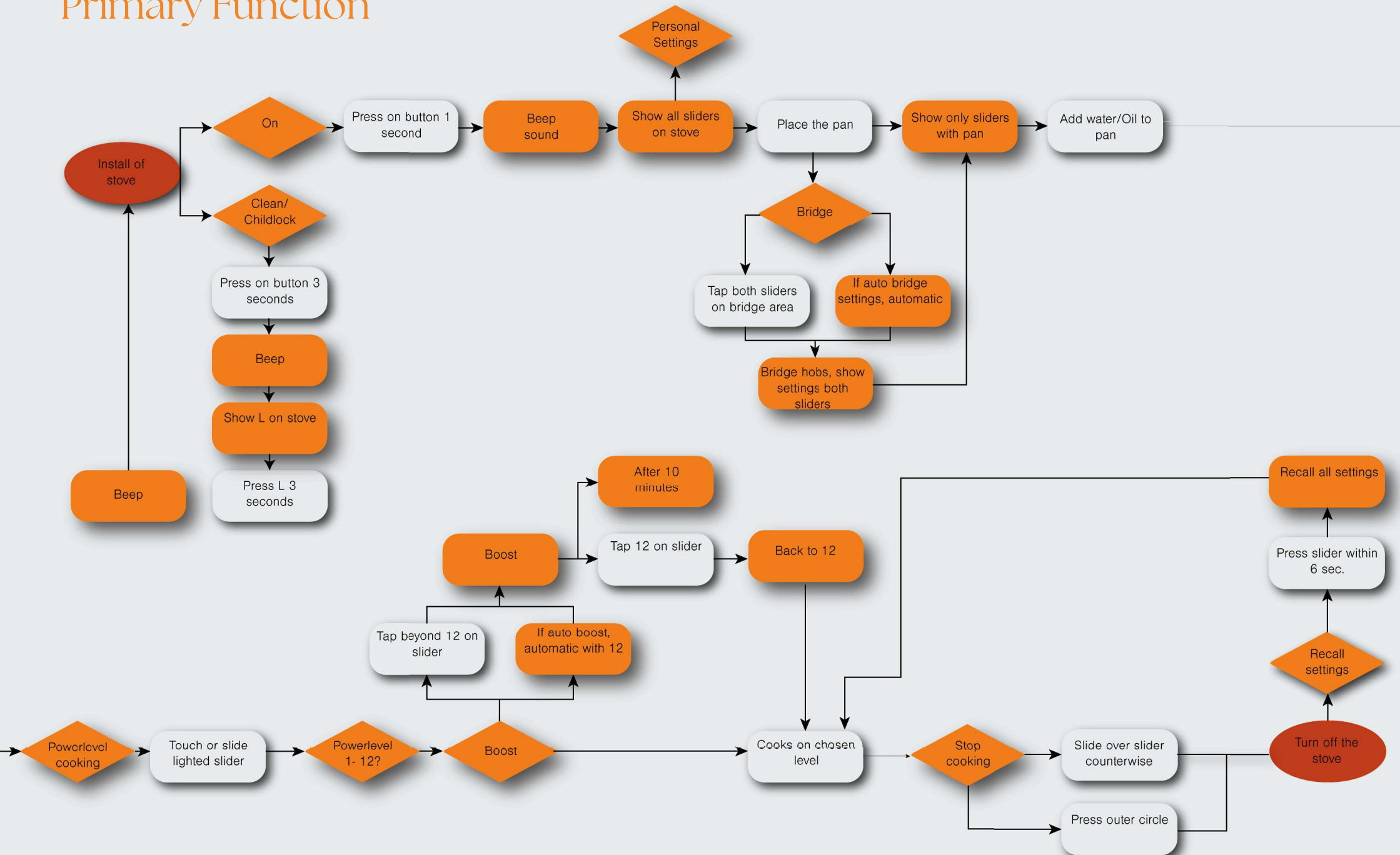
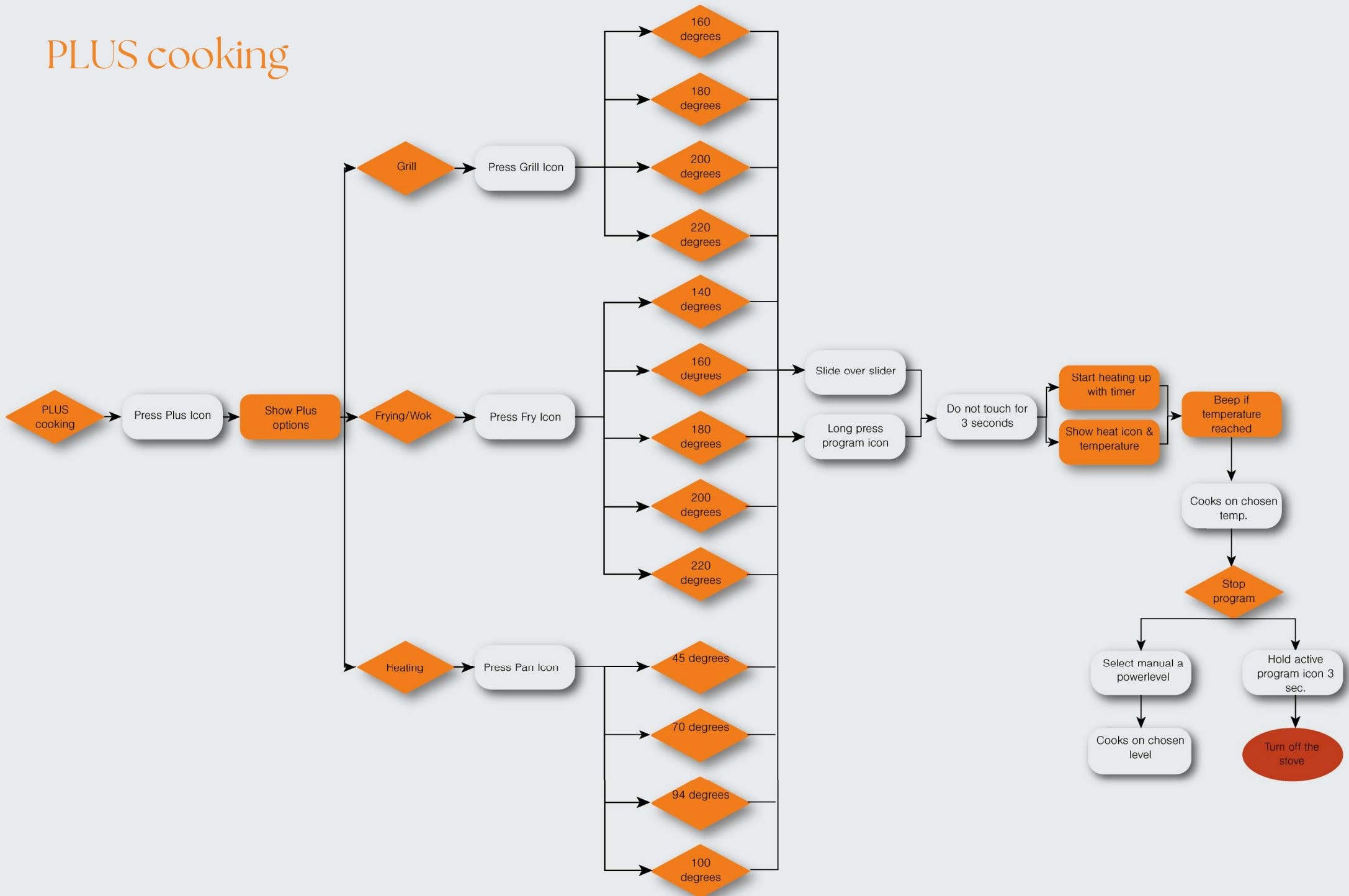


Figure 7: Overview of Functions.

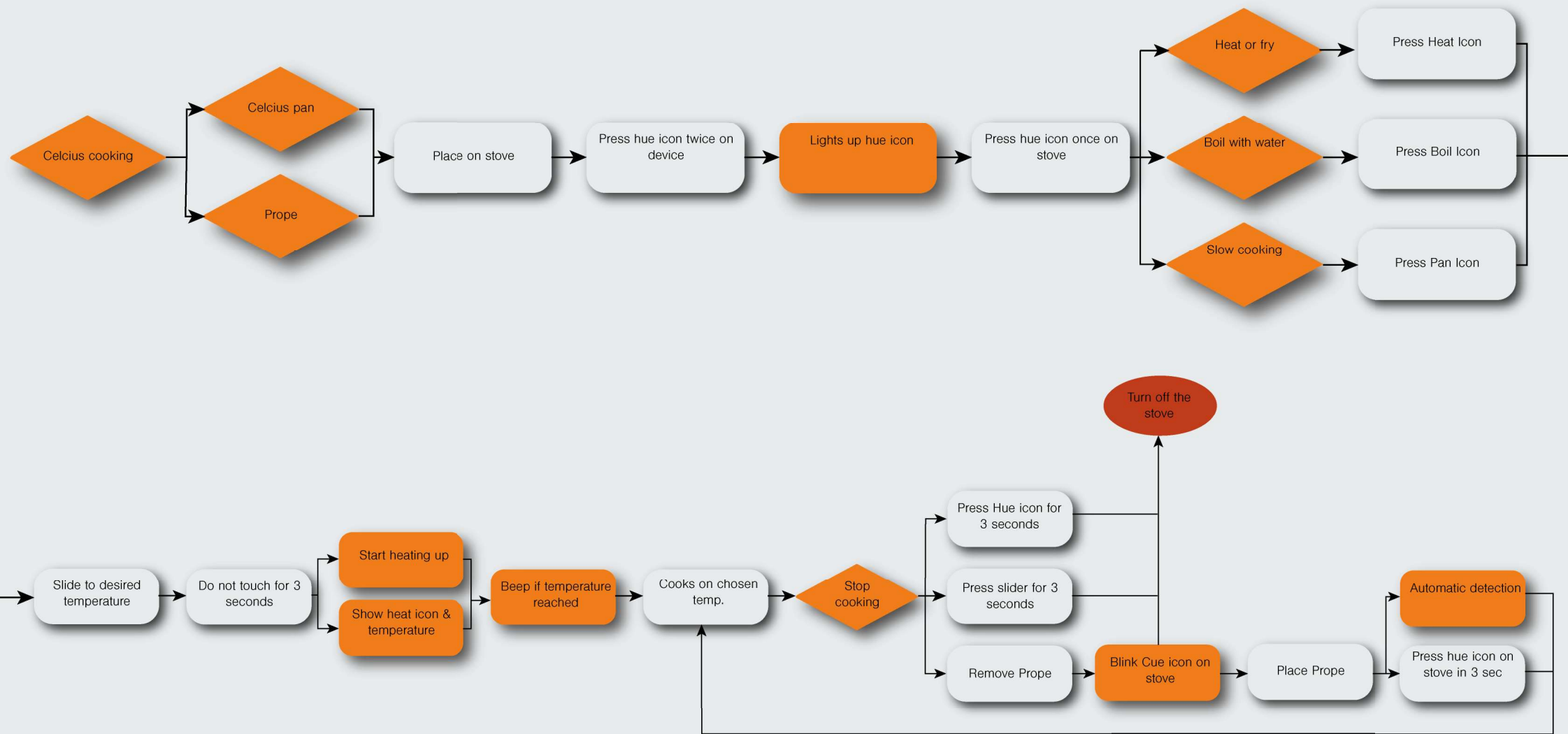
Primary Function



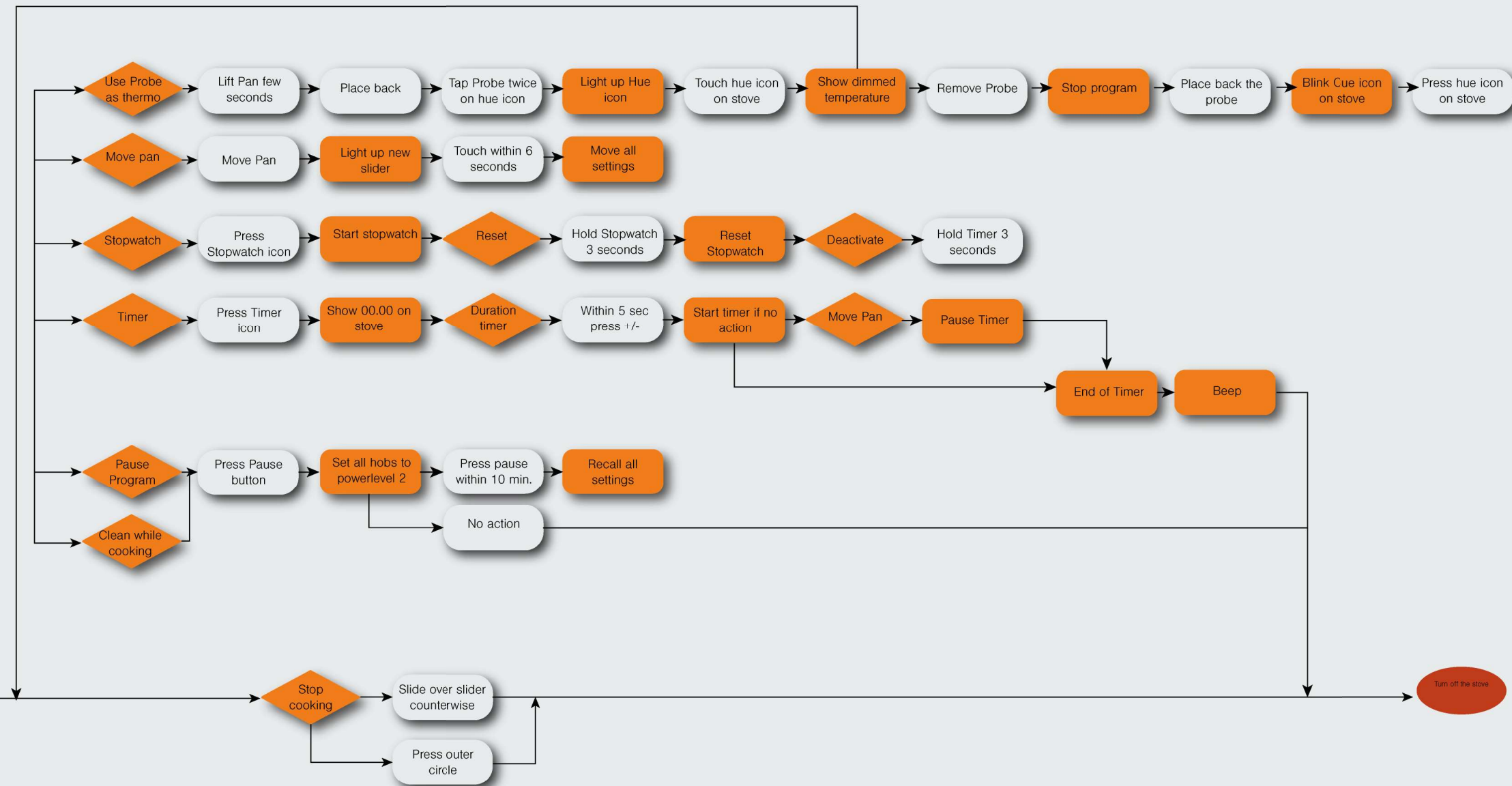
PLUS cooking



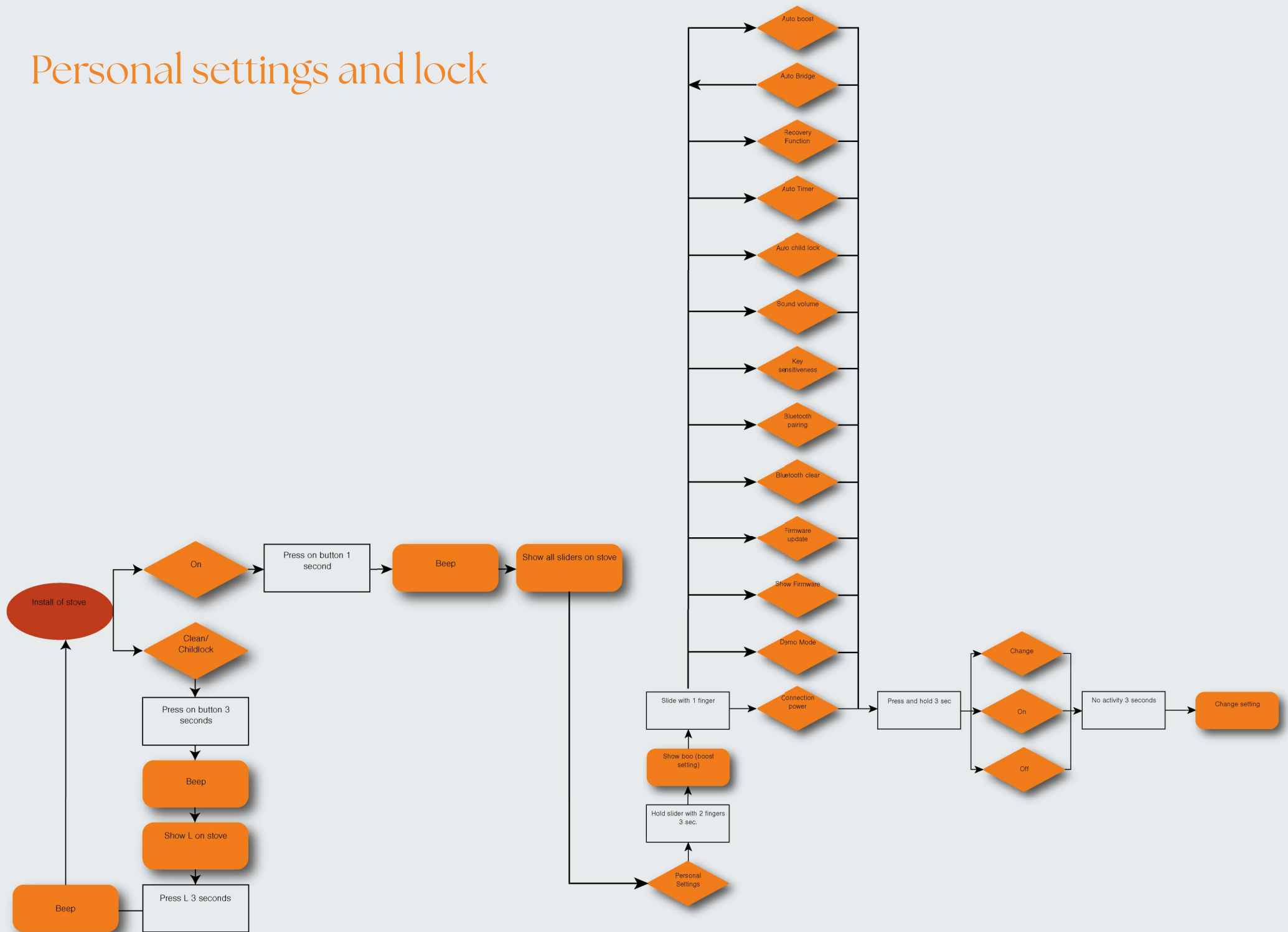
Celcius Cooking



Extra functions



Personal settings and lock



Users

According to Atag, there is not one chef. They aim their products at three main groups:

- The social chef: someone who likes to cook for an audience.
- The semi pro: someone who cooks to improve their own skills.
- The pleasure seeker: someone who likes trying out new things, but not in complete detail.

Atag is a premium brand with high-priced products aimed at a high-class user group. It's important to design for secondary users, such as family, friends, or chefs, since the primary users may not cook themselves.

Based on these target groups, three personas are created. These persona's have different motivations to buy the product and would therefore also use it in a different manner.

Daniel Hastings



Gender : Male
Age : 35
Education : secondary vocational education
Occupation : own company in IT

Biography

Thijs is a 35 year old who is married to Olivia. They have 2 kids, aged 9 and 13.

Motivations

Both Olivia and Thijs enjoy cooking. During the weekends they invite friends over for dinner and drinks. Together they try new recipes. He likes innovative new products and loves to combine that with cooking.

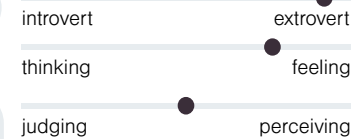
Goals

- try new recipes for an audience
- have the newest appliances for cooking
- combine cooking and technology for the best results

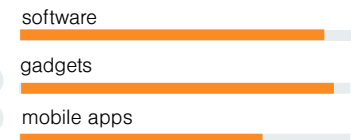
Frustrations

- some recipes require temperatures that are hard to get right with a hob
- wants more technology gadgets in the kitchen so he can take cooking to a new level

Personality



Technology



Brands



Olivia Hastings



Gender : Female
Age : 33
Education : Master's degree
Occupation : dentist

Biography

Olivia is a 33 year old who is married to Thijs. She has 2 kids, aged 9 and 13.

Motivations

Olivia and her husband enjoy cooking. In the weekend they invite friends over for dinner and drinks. She then likes to try new recipes. When she comes home from work she wants a meal that is fast but healthy.

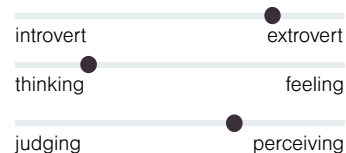
Goals

- try new recipes for an audience
- cooking can be easy and complex
- cook healthy and tasty food

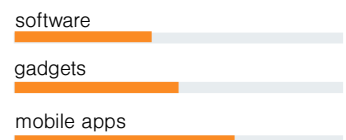
Frustrations

- some recipes require temperatures that are hard to get right with a hob
- current cooker is outdated, her friends have newer products with more options

Personality



Technology



Brands



Ella Jones



Gender : Female
Age : 24
Education : student tourism
Occupation : babysitter

Biography

Ella is 24. Next to her study she looks after the kids of the family Cromhout.

Motivations

She enjoys taking care of the kids because they listen well. Olivia and Thijs asked her to make them a healthy dinner, if they are out for the night. Ella loves different cuisines.

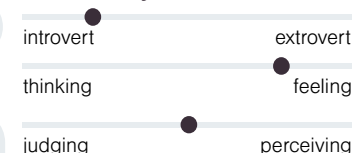
Goals

- try new cuisines
- cook healthy and quick meals
- keep the kitchen safe with kids around

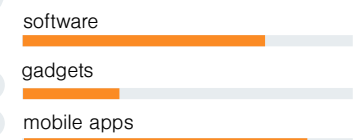
Frustrations

- She has to have eyes everywhere, she needs to keep an eye on the kids and also cook at the same time.

Personality



Technology



Brands



Context of use

Onion of use

Figure 8 maps out the context of use. The map describes the user, environment, other user and symbiotic.

Further explanation of the mapping can be found in Appendix B: Onion of Use.



User

Environment

Other User

Symbiotic

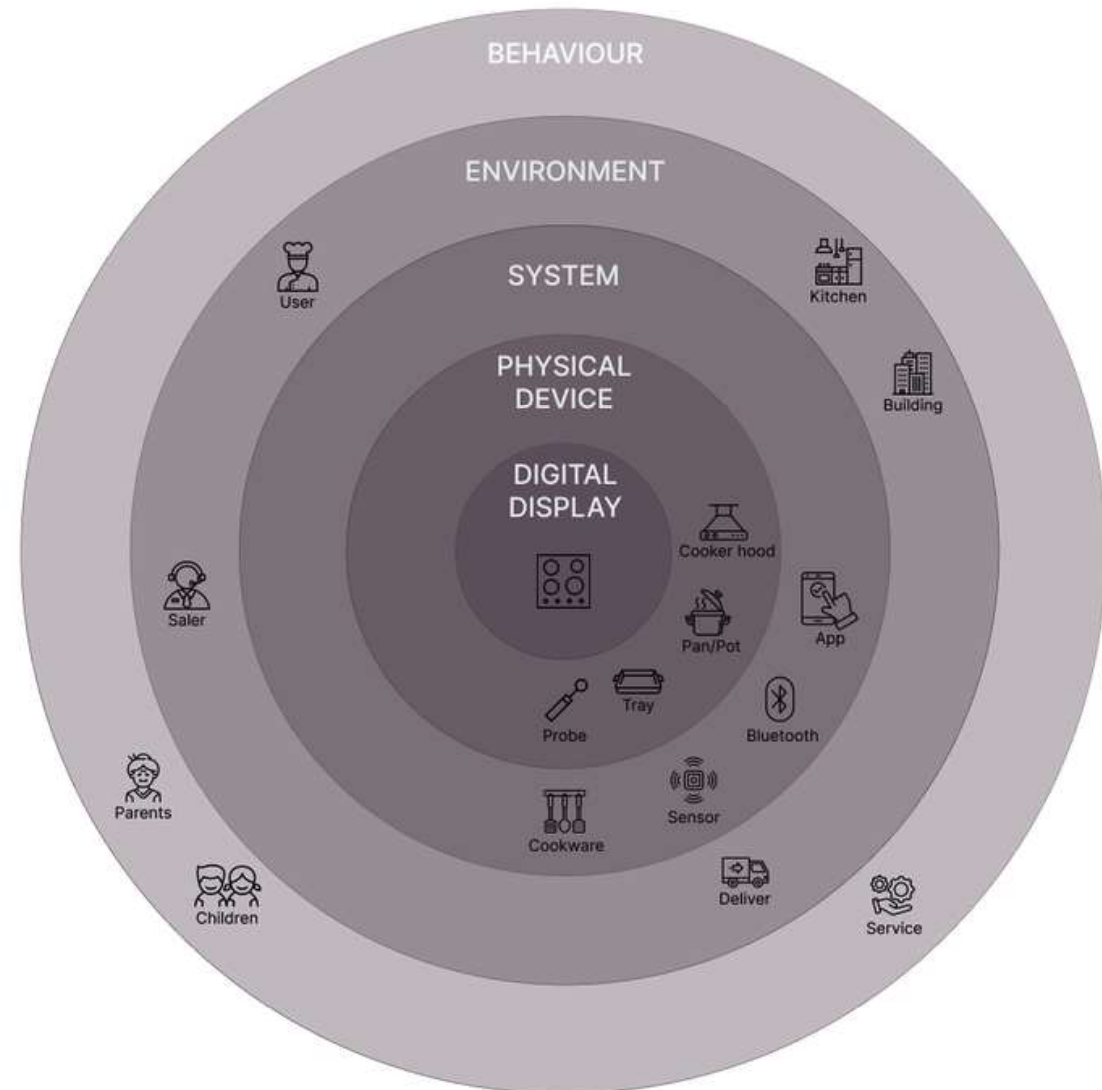


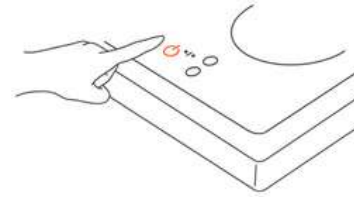
Figure 8: Onion of Use.

Storyboard

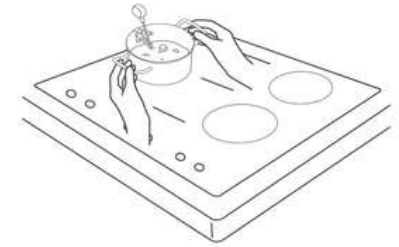
Figure 9 depicts the ideal scenario of a user using an Atag induction hob at home.



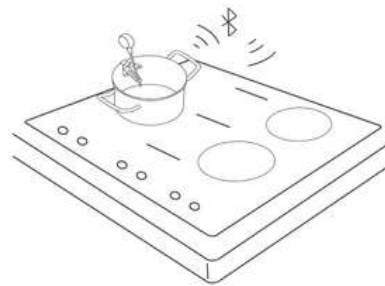
1. Assembling the probe and pan/pot together



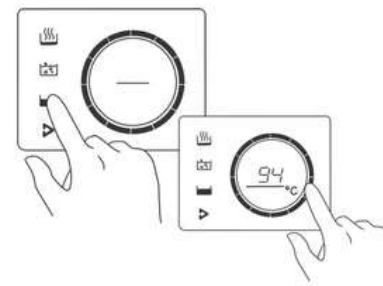
2. Activate the induction hob using the touch screen



3. Place the assembled pan/pot on one of the cooking area



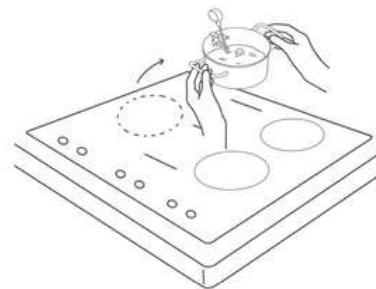
5. After the probe connect to the Induction hob, Hestan Cue mode can be choose



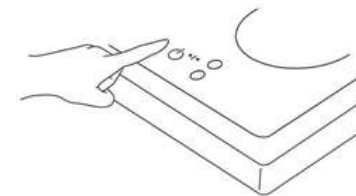
4. Find the lit touch screen and adjust to desired cooking mode



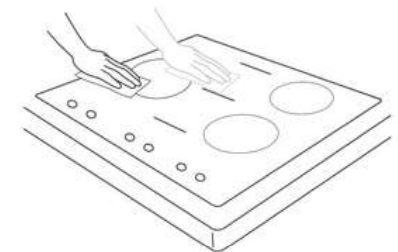
5. Various values can also be customised using the mobile app



6. When cooking is complete, take the pan/pot away



7. Inactivating the induction cooker with touchscreen



8. Cleaning the induction cooker

Figure 9: Scenario sketch of use flow.

2 ■ Quality of use

This chapter we focuses on our approach to user testing, where we combine forehead assessments, user tests, and questionnaires to understand user experiences deeply. This phase is crucial for identifying user interactions, preferences, and challenges with our design, guiding us towards actionable insights for improvement.

First hand experience
Second manual test
Analysis of reviews
Questionnaire
User test



First Hand Experience

Appliance

All five team members conducted a hands-on experience with the Induction Hob. We documented our impressions in a form and analysed the results to identify the initial pain points and strengths of the appliance shown in Table 1. The detailed results are available in Appendix C.

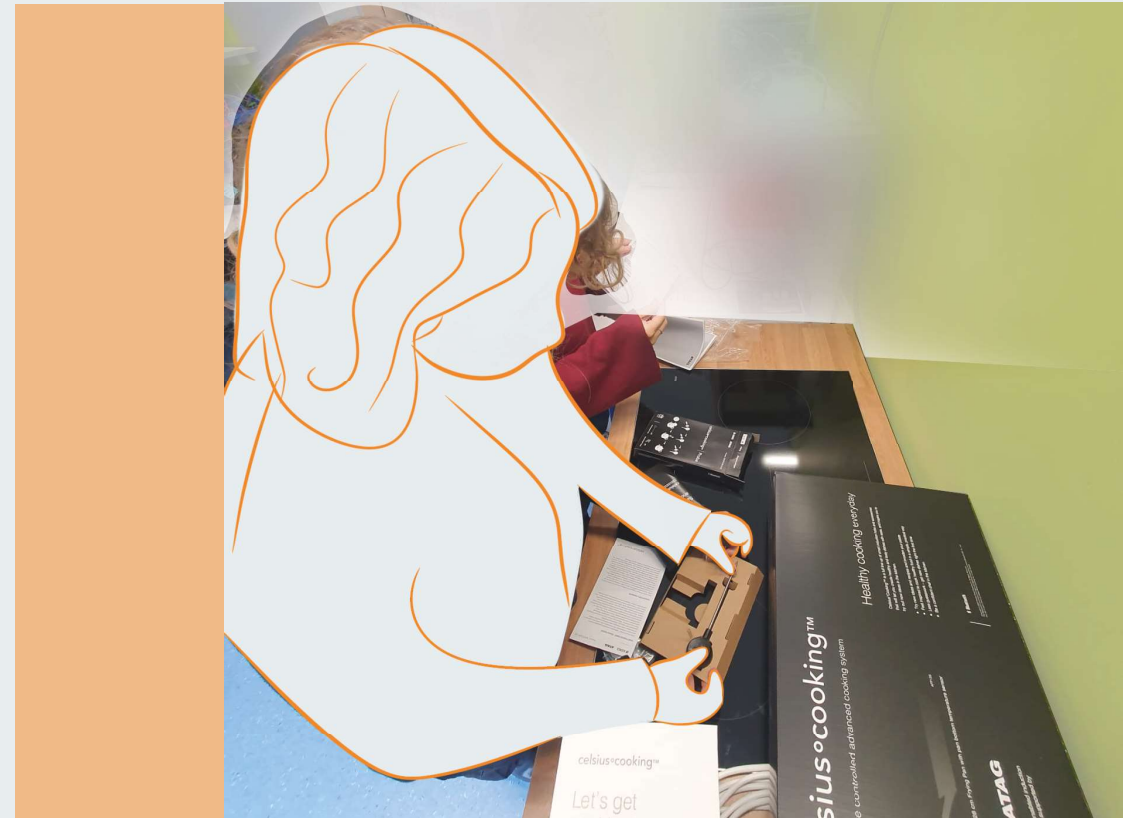
Objectives

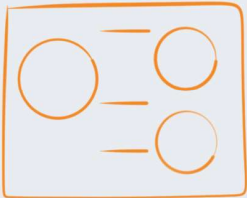


- Identify the product's components
- Document everyone's first impressions of the components
- Have first product interaction

Packaging & attributes

The Product & Accessoires



	Component	Action	Emotion	Reason
    	Hob - Surface	Using the clean and new glass surface	Pleasure	Due to the novelty, feel and premium aesthetic of the appliance
	Hob - Surface	Moving the pan from one cooking zone to another to test the ' pan detection '	Anxiety	Due to error, misuse or damage because of the lack of understanding.
	Hob - Cooking zones	Using the Bridge zone	Confusion	It is a new function, and the hob's interface wasn't responding
	Hob - Symbols (Visual Cues)	- Observing the heat symbol displayed for when the cooking zone was still hot - Responding to the blinking of the timer and stopwatch functions	Appreciation	Ease of understanding the function due to the appropriate response and timing of the visual cues <i>Note: Importance of using symbols that are universally recognized (like the waves symbol for heat)</i>
	Hob - Functionalities	Trying out the many functions visible	Confusion	While the range of functionalities was initially appreciated, the confusion arising from not knowing how to use them properly highlighted the need for time to adapt to new functions and technology
	Pan and Probe	Attaching the clamp and probe on the pan's edge	Discomfort	Due to the sound of the metal surfaces scratching against each other

Analysis of reviews

We reviewed user feedback on Atag's product and comparable brands in the Netherlands. Despite the product's novelty leading to few reviews, we identified key issues based on user experiences (see Figure 10).

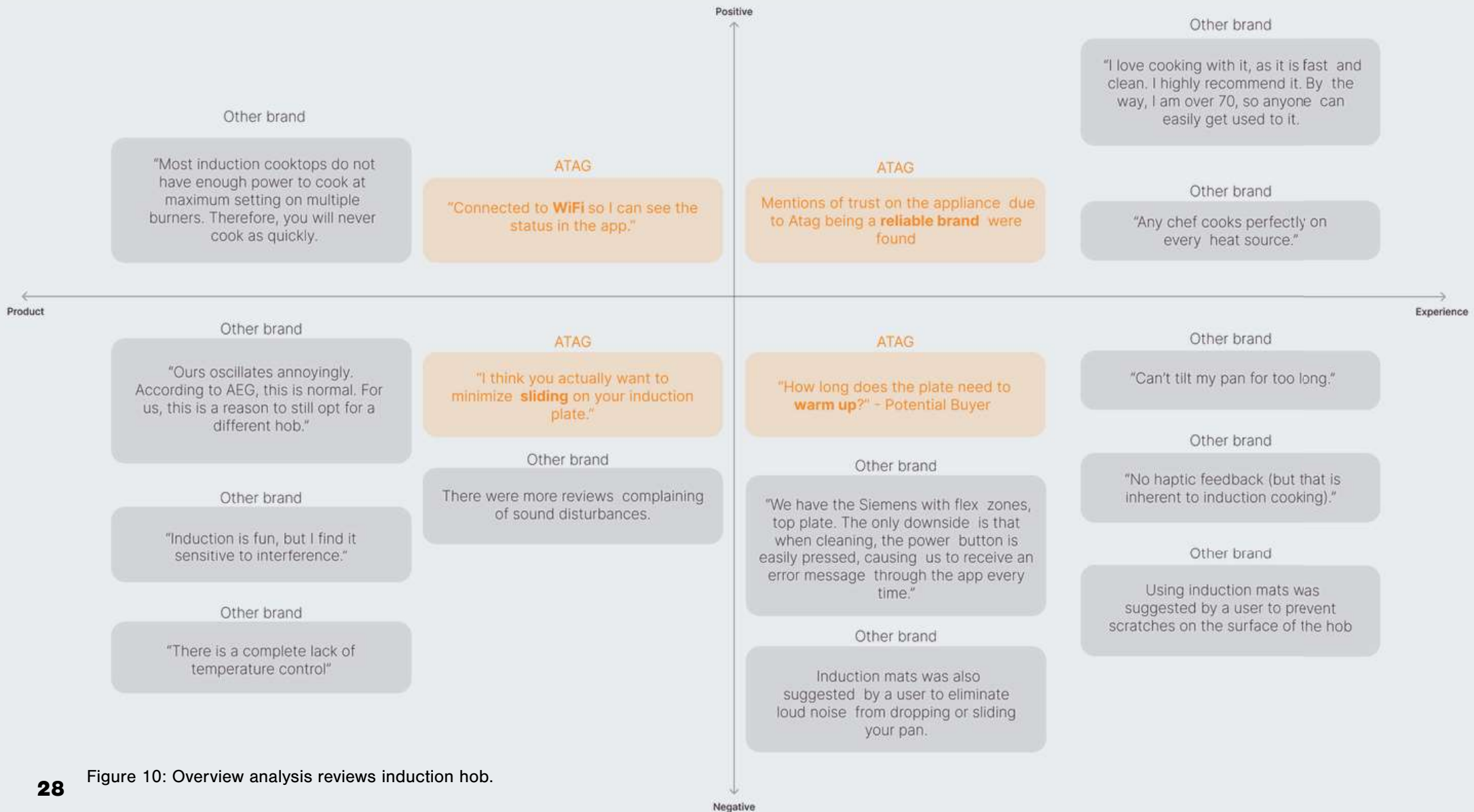
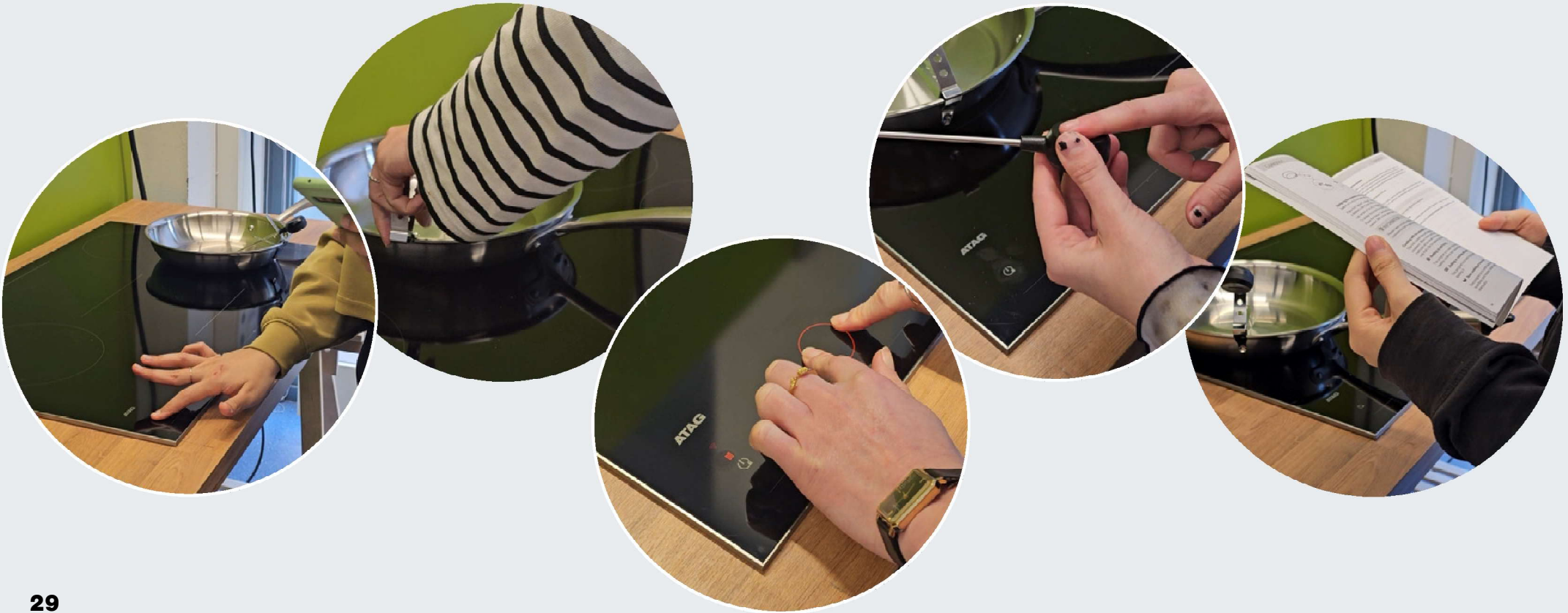


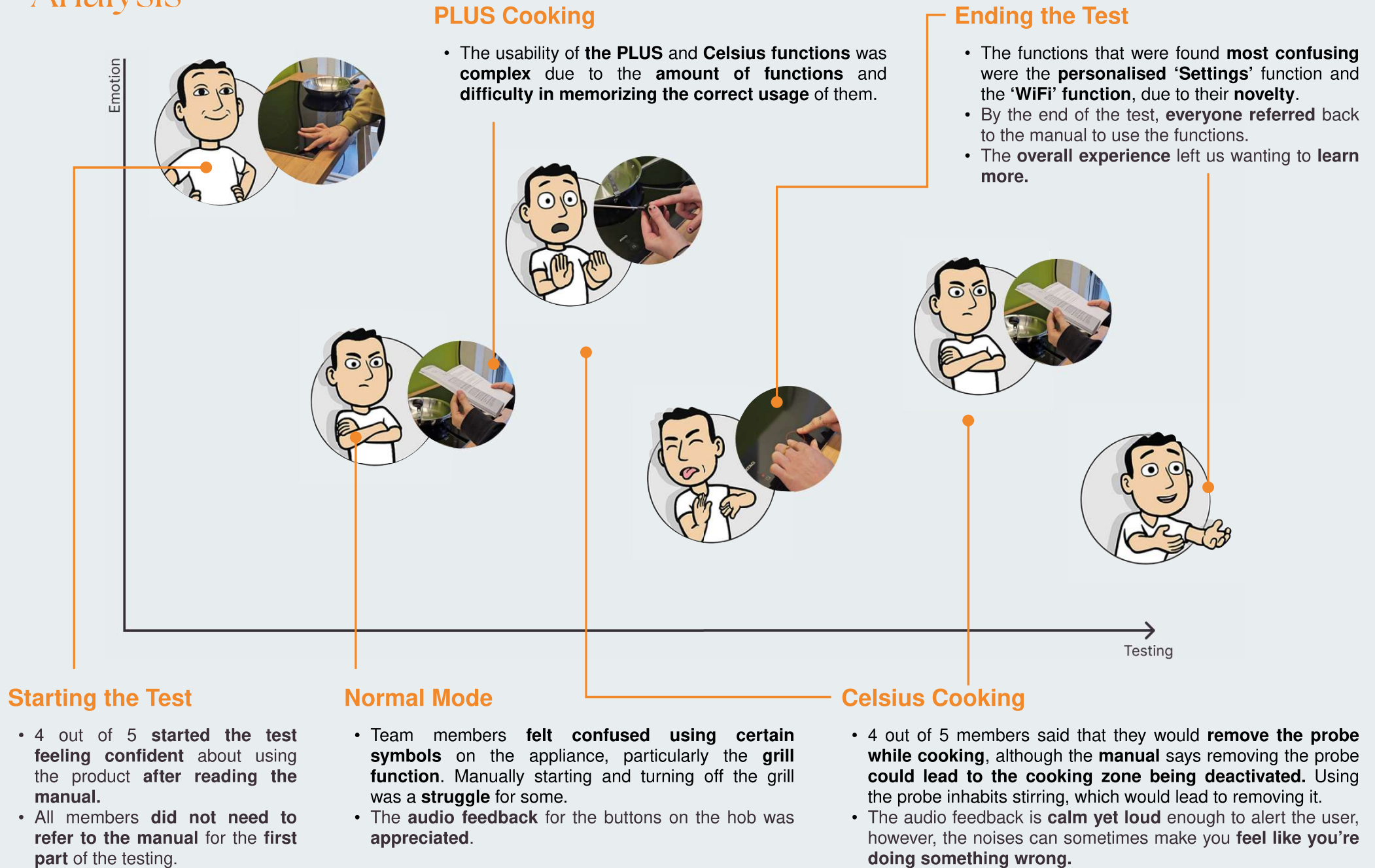
Figure 10: Overview analysis reviews induction hob.

Cognitive walkthrough

We conducted a second round of testing after thoroughly reading the manual. We aimed to inspect the usability of the appliance, document emotions felt while using all functions, uncover problems faced, test memorability of functionalities, and test dependency on the manual. Each member documented their process and emotions through a form, with only one other member present during testing to moderate the discussion and take videos.



Analysis



Questionnaire

We administered a questionnaire to 33 participants and targeted only those who cook more than once a week. We also attempted to differentiate the results based on the experience levels of the respondents. However, since there wasn't much variation in the answers, we decided not to pursue this analysis any further.



33
participants



Improving cooking skills from the top 4 mentioned:

1. Learning from friends and family and by doing
2. Online Video content & Social Media
3. Cookbooks
4. Investing in new cooking appliances



60.6% have been regularly cooking for more than 4 years, out of which **15.2%** have been cooking for 15 years.
39.3% have recently become regulars



67.6%

Cook 2 to 4 times per week



20.6%

Cook 5 to 6 times per week



8.8%

Cook everyday

Analysis

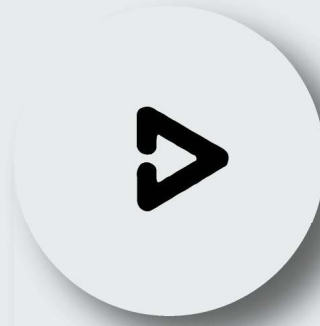
We conducted a survey (see Appendix E) to determine if potential users could intuitively understand the symbols present on the hob. The survey was divided into three parts. The first part tested the participants without any context. The second part tested them with context, and the third part assessed their intuitive interaction in turning the hob on and off.



75% were right
14.6% were close
10.4% no clue



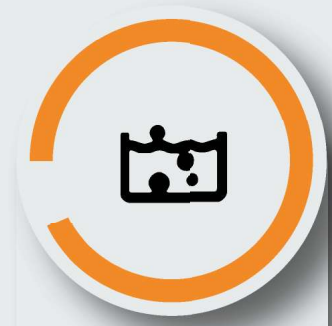
18.5% were right
63% were close
18.5% no clue



0% were right
0% were close
100% no clue



0% were right
8.5% were close
91.5% no clue



93.8% were right
0% were close
6.2% no clue



0% were right
55.9% were close
44.1% no clue



13.9% were right
0% were close
86.1% no clue



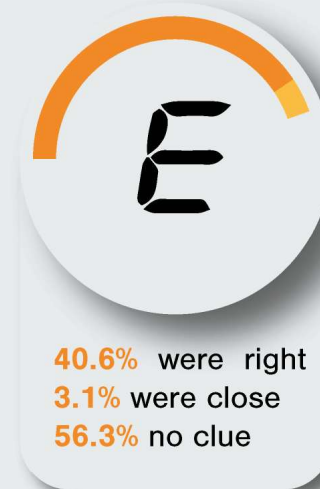
20% were right
65.7% were close
14.3% no clue



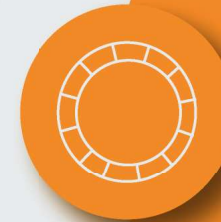
0% were right
9% were close
91% no clue



0% were right
0% were close
100% no clue



In context



73%

said this would cook normally with levels from 1-12, some mentioned the boost function would go here.



52%

indicated this is more advanced cooking, some signified it is for what kind of pan used or special programs for specific dishes



33%

said it would be for cooking with precise temperature but only mentioned it while making things that require temperature. The rest had no clue, or would not use it.

Cooking settings



Users associated the icon with cooking and specific cookware. Responses included boiling water, simmering, and heating pots or pans. Some participants did not understand.



Participants associated the icon with high-heat cooking activities like frying, baking, and broiling in a frying pan. Some mentioned cooking pancakes, eggs, and meat. There were fewer unsure answers.



Only 18% of respondents understood the meaning of the icon related to heating or cooking on a surface with a grill. Some expressed confusion or had no clue about the icon's meaning.



Responses suggested gently heating liquids or food on low temperature without boiling them. The theme was low or gentle heat, with some mentioning temperatures below 100 C. However, a few didn't understand.



This icon was interpreted as something used for boiling water or liquids. A few explicitly mentioned heating until the liquid boils at 100 C. The overall theme seems to be cooking at high temperatures. There seems to be a clear consensus.



Participants had different associations with the icon, including steam-related actions or maintaining a temperature. Some mentioned specific cooking activities, while others were uncertain about its meaning.

Intuitive interaction



Turning the hob off

33% participants (most common) suggested swiping the circle to 0. Other participants mentioned:

- Clicking the highlighted pan icon (in reality, it is a long press)
- Long press where the temperature is shown
- Look for the on/off button
- Click anywhere on the platform to exit

Diverse responses may indicate confusion or uncertainty about the interface's functionality, suggesting room for interface improvement or more evident design cues.



Turning the hob off

30% participants (most common) indicated again swiping to 0. Others mentioned:

- Press/long press the number (18% of responses)
- Tap on boost (18% of responses)
- Long press the plus icon
- Press the square

Here the intuitive part is swiping counter-clockwise, yet there is still uncertainty. There was also experimental behavior, with some pressing other buttons on the screen.



Turning the hob on

60% (most common) participants, although incorrect, noted to press the "+" icon. Other responses included:

- Press/long press the "-" icon in the centre (21%)
- Swiping around the circle "as if I were twisting a knob" (1 person, but interesting since this was the design intention).

User Test

Our team conducted a series of three comprehensive user tests to evaluate the strengths and weaknesses of the product and pinpoint the key areas for improvement based on user interaction. This chapter elaborates on the test preparation process, provides participant details, and presents an in-depth analysis of our findings.

Research Questions

To establish our test plan, we first needed to define what we wanted to achieve from it. Therefore, we developed the following research questions:

1. Can the user complete the interaction process within the hob's primary function?
 - Is it user-friendly?
 - Can they operate it with ease and without any hesitation?
 - Can they use the appliance without the manual?
2. Can the user cook with ease using the hob's primary functions, secondary functions, plus cooking, and Celsius cooking?
3. Is the user willing to learn or adapt to the new functions?
4. Does the cooking hob's experience empower the user?
 - Does it boost their confidence?
 - Does it arouse curiosity or frustration?
5. Does the user feel like the appliance could help them improve their cooking skills?
6. Which part of the user journey's interaction was the most confusing or frustrating and prompted the most questions?

Context of use

We conducted a user test in the Applied Labs and aimed to make it more realistic by creating a homely cooking scenario. We provided them with all the necessary cues and prime them in the scene. To enhance authenticity, we displayed an array of spices, a cutting board, different pans, oil, and plates with cutlery for serving.



Test Plan

We narrowed the focus of our user test on the hob to daily tasks and interactions. Our aim was to identify design opportunities that would enhance users' daily use of the machine. We tested the primary functions of the timer/stopwatch, cooking with power level, plus level, and Celsius cooking. We also completed the entire user flow, which included cleaning, preparing, and finishing. See the next page for the specific scenario's. We briefed the participant before the test and used a template for the observer and a moderator script to keep coherence in the test (refer to Appendix F & G for detailed scripts). After the test, we wrapped up with the participants in a casual conversation to ensure that we did not miss out on any valuable insights shared in a conversation outside the test setting.

Participants

We conducted a user test with three participants who were all regular cooks and had experience cooking with an induction hob. We made sure to recruit participants with prior experience to avoid any initial learning curves and they had an equal ability to cook. This allowed us to have a balanced group with similar profiles.



Expertise level: 8 out of 10

Type of cook: Participant 1 is in a cooking club, here he likes to experiment with different new functions and recipes. He does not follow the recipes strictly.

How often? Cooks 3-4 times a week

First steps for using a new cooking hob: He would first start up slowly and start figuring out by him self. Whenever he is stuck he will look at the manual.



Expertise level: 7 out of 10

Type of cook: Participant 2 likes to cook for guests and test out new recipes but does not always have the time to spend a lot of time in the kitchen.

How often? Cooks 5-6 times a week

First steps for using a new cooking hob: She would start playing around with it, without reading the manual.



Expertise level: 6 out of 10

Type of cook: Participant 3 likes to try new recipes, cooks a lot of Indian food.

How often? Cooks 5-6 times a week

First steps for using a new cooking hob: He would first read the manual, but does not like to read a lot.

1

Scenario: At their usual cooking time, starting the cooking process after having prepared all the ingredients and utensils.

Tasks: Turning on the hob and putting the pan in place.

Objective: To evaluate the hob's initial user-friendliness, assess the effectiveness of the indication lights, and gauge users' first impressions.

2

Scenario: Prepare and boil the rice for the dahl.

Tasks: Use the hob to set the timer and power levels for cooking the rice.

Objective: This task is designed to examine the user's initial approach when asked to perform a simple task such as boiling rice. Observe their interaction and determine if it is a positive experience.

3

Scenario: Sauté onions for the dahl using the special function on the hob.

Tasks: Fry onions with additions while setting up the fry setting of the PLUS cooking settings, see the preheat function, wait for the skillet to warm up, and fry the ingredients subsequently.

Objective: To evaluate the ease of use and likability of the PLUS function, as well as the comprehension of the preheat function.

4

Scenario: Simmer the dahl at an exact temperature.

Tasks: Set up either the probe or the skillet to simmer at exact 90 degrees and set a timer after the skillet has reached the right temperature.

Objective: Observe the use flow of the Celsius Cooking and determine if this additional function is well-received, as well as the comprehension of the two alternating temperatures when warming up.

5

Scenario: Cleaning the stove after use.

Tasks: Explore the lock function on the hob.

Objective: To observe the user flow of cleaning the induction hob and whether one would want to use the lock function for this.

Discussion

We attempted to establish uniform testing conditions, but unforeseen factors were observed. Participant 2 has a cooking hob equipped with a preheat function. This familiarity enabled her to identify the icon and process better, leading to less difficulty with recognition in that specific area. Furthermore, while Participant 3 was being tested, the hob was in Lock mode, which resulted in a distinct initial interaction and affected the cleaning scenario.

Data Analysis Plan

The observer filled an observer form (Appendix G) during user tests, gathering thorough notes. To analyse the tests precisely, the data was categorized based on specific terms:

To understand the product's pain points comprehensively, we generate various graphs that display the experience of each task and participant. Additionally, a graph demonstrating the emotions of each participant during product usage provides valuable insights into areas that require improvement (see Appendix I&J).

Objective: What does the user have to do?

Effectiveness: Was the user able to complete the task? This is measured on a scale of easy, medium and hard.

Errors: What difficulties did they encounter?

Satisfaction: How satisfied did they feel with the task and product in that task?

Efficiency: How long did it take to complete the task?

Experience: What was the overall experience with the product during this task? This is measured on a scale of positive, neutral and negative.

Quotes: What did the participant say out loud during this task that is of importance?

Other: Are there any other things that stood out during this task that are of importance?

User test analysis: Strong & Pain Points

The strong and pain points from each user test were clustered and compared (Appendix H) in order to summarize the main insights as listed below:

😊 Strong points:

- + **Placement of pans** over cooking zones was **easy**.
- + Users **were curious** to explore various functions and found the **timer, power levels**, and certain **presets (PLUS mode)** to be **useful**.
- + **Cleaning** was found to be **easy**.
- + It was **clear** when the **hob was still hot**, due to the icon.)))
- + It was **easy** to **change settings**, (once found in manual) and was found to be a **nice option**.
- + The **countdown timer** function was **understood** and **appreciated**.
- + **Audio** feedback of the **timer** was **enjoyed** by all users.
- **Normal Mode (1-12 power levels)**
- + Most functions here were **intuitive**. The interaction with the **Iris slider** was compared to that of **turning a knob** (as intended by Atag).
- + **Audio** heard after the **pre-heating reached completion** was also **enjoyed** by all users.

😞 Pain points

- **Pan detection** responded quite **slow**.
- **Letter symbols** on the hob were found to be **confusing**. (L for Lock, E for Error, H.M. for Hours and Minutes)
- **Confusion** due to the **design of buttons** left users asking **where the buttons were** or **how to proceed**.
- Users were **unclear** about what the **stopwatch** icon represented and **didn't find it intuitive**.
- Some users assumed that the hob would **turn off** at the **end of the preheating timer**.
- **Sounds** during **errors stressed** the users. There was a combination of a **loud beeping**, no **clarity** on the **error type**, which **prompted** the users to **shut the hob off**. For example:
 - when the probe touched the pan
 - when liquid spilled on the hob
- **Too many steps** were taken to **set up** the **hob**, which **delayed the cooking process**.
- **Some participants** didn't know of the **lock function** for **cleaning**, **neither** was it **clear** if it was **on/off**. They **strongly felt** that it was **not necessary**.
- The **settings** cannot be **accessed** with **pans on the hob**. This was **unclear** to users.

☹ Pain points (contd...)

● The Manual

+ It was **confusing** and **topics** were **not easy or fast to navigate while cooking**. **Tasks** were **not easier** after reading it.

● Normal Mode (1-12 power levels)

+ Users **did not realize** the **difference** between 'Normal' mode and 'PLUS' mode.

● PLUS Cooking (presets)

+ 140° for the **frying mode** was **too hot**. Users **could not decrease** the temperature to avoid burning the food and instead, removed the pan from the hob.

+ **Not clear** of which **temperatures** are to be **used** for which cooking technique.

+ Users were **frustrated** due to a **lack of control** when the functions got **locked**.

+ This **timer** could be increased by 1 minute but decreased by 10 seconds, which was **confusing**.

+ It is **difficult** to **get out** of the **PLUS** settings. Users need to **double tap or long press** which is **not intuitive**.

● Celsius Cooking (°)

+ The '**pause**' function **did not work** here.

- Celsius cooking does not remember your previous settings when you exit accidentally. This, then, requires the user to reset everything which takes extra time making it very inconvenient.

- **Problems with connection** of the **probe and/or pan** to the hob left users **feeling unsure**, making them use buttons on the hob to connect (not as intended).

- It was **difficult** to see if the **probe** is **1cm** above the pan's surface, especially with some food in the pan.

- Major **fluctuations** in the **precise temperature** displayed on the hob shows **unsteady** Bluetooth **connection**.

- Most users **preferred** to use the **normal** or **PLUS** mode rather than this one.



User test analysis: UEQ questionnaire

During the user tests the participants were also asked to fill out the short version of the UEQ questionnaire after every task. This short version asks users to judge the product on the following aspects:

Obstructive vs. Supportive

Complicated vs. Easy

Inefficient vs. Efficient

Confusing vs. Clear

Boring vs. Exciting

Not interesting vs. Interesting

Conventional vs. Inventive

Usual vs. Leading edge

The questions in the UEQ consist of two dimensions:

Pragmatic Quality: This demonstrates the usability of the product.

Hedonic Quality: This demonstrates the aspects of a user interface that appeal to a person's preferences.

The questionnaire results are visualised per task, which can be seen in Appendix K & L, and the average results can be seen in Figure 11. It can be concluded that Celsius cooking and changing the settings is rated the lowest for pragmatic qualities. The participants experienced these functions as complicated and confusing. In the hedonic attributes, there is room for improvement in the product's cleaning. Likely because it is seen as a common task, no different from other products. It is important to note that this issue is less severe than the issues found in the pragmatic qualities; it is only slightly negative and is more the overall objective than a product issue.

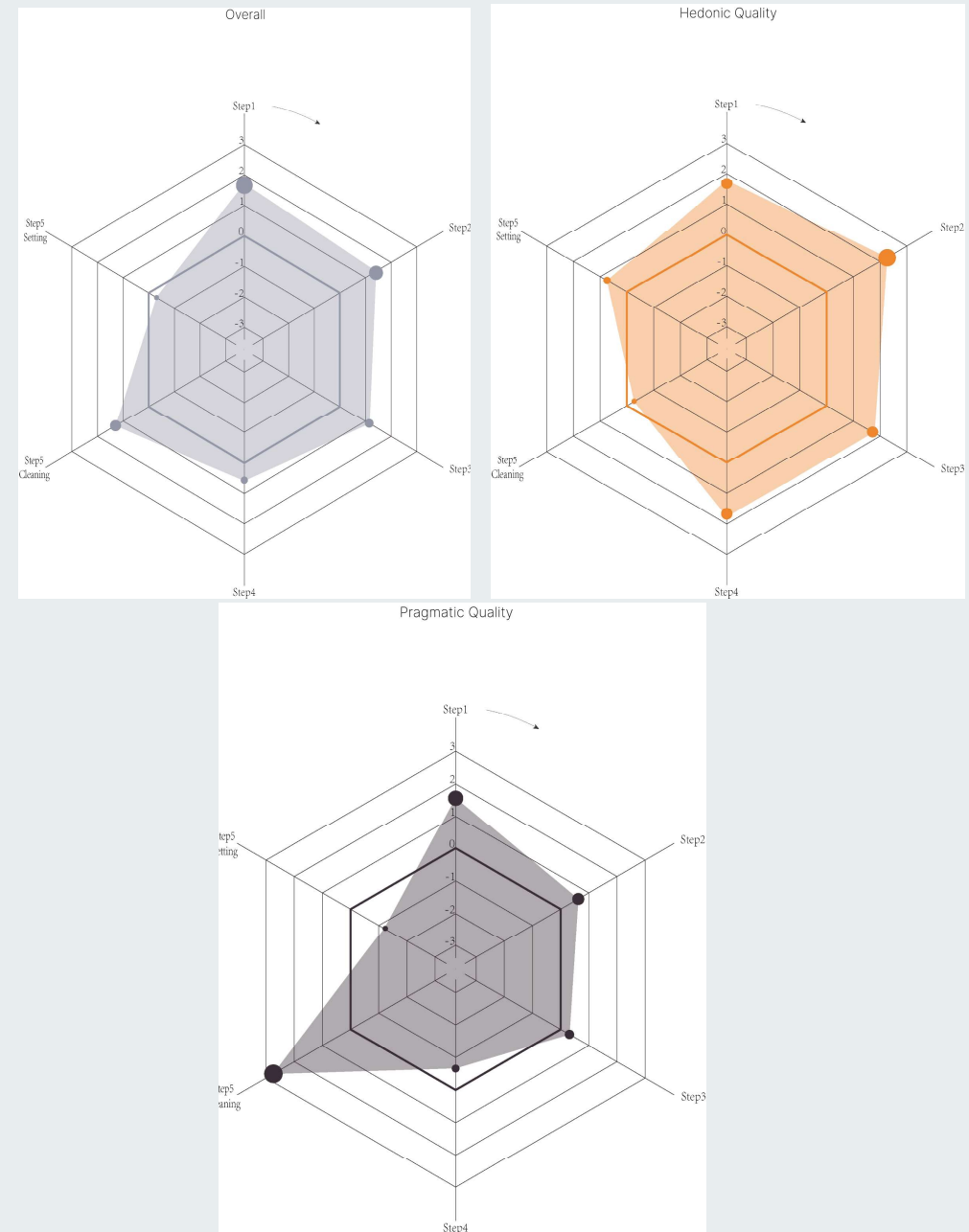


Figure 11: Analysis UEQ

“

What are you telling me, I can cook?
You don't have to do it for me.

”

“

You don't want the hob to pause by itself, you want to stir while cooking.

”

“

If the hob displays a number I should be able to increase or reduce it.

”

“

It says now you're ready but maybe I'm not (About heatup function).

”

“

I don't think I used it in the right way.

”

3 ■ Problem definition

This chapter delves into the problem statement identified through user testing, pinpointing the main challenges faced by users and outlining the desired interactions to address these issues.

Problem statement
Current and desired
interaction



Problem statement

Through the analysis performed in this report, we identified the problem statement that guides us towards the redesign scope. The usability test revealed that the hob is too complicated and its complexity overshadows its functionality. Users felt a **lack of control and found it challenging** to switch between cooking functions, which made them feel insecure during the cooking process. Although the PLUS cooking zones were intended to provide easy-to-follow programs, they were too complicated and restrictive, **taking away the user's freedom** to cook as they wish.

One of the significant issues with the product is **onboarding**, as it has a steep learning curve. The manual is often confusing, and users don't have enough knowledge to cook with degrees, resulting in burnt food. Furthermore, users found the feedback provided during product use to be unclear, making it for example challenging to understand errors or when the product was locked in preheat mode. This feedback was often accompanied by confusing and harsh sounds, leading to users **feeling overwhelmed, confused, and frustrated**. These emotions can reduce the joy of using a new advanced cooking appliance, which is intended to elevate the cooking experience, not diminish it.

Complicated interface

Feeling challenged, insecure, and restricted

Lack of control and freedom

Steep learning curve and unclear guidance

Confusing product feedback & alerts

Lack of control due to locked functions

Complexity overshadowing functionality

Current & Desired interaction

Current interaction

According to the problem statement and the pain points, there are several main issues with the Atag Induction Hob. The steep learning curve has diminished the excitement of getting a new product and experimenting with it. As a result, users have been left feeling confused and frustrated by the current interaction, despite finding the product fancy and enjoying most of its functions. They are open to learning more, provided that it is accessible and easy to understand without feeling incompetent.

Desired interaction

Users are usually eager to learn, as long as the learning process doesn't make them feel incompetent or overwhelmed. To tackle this issue, we need to focus on creating a new interaction that guides users, boosts their confidence, and increases their excitement to experiment and improve their home chef skills.

Solution focus

The redesign should aim to be simpler, with a guiding interaction that provides a more informative and responsive interface. The new design should address these fundamental usability issues and enhance the overall cooking experience by providing a streamlined, simplified, and intuitive interaction that is free from any unnecessary complexity. Allowing users to enjoy the process of cooking and envolving their cooking skills with the support of the hob.



Confusing

Frustrating

Lack of Control

Overwhelming

Guiding

Intuitive over time

Confidence boosting

Evoke eagerness

4 ■ Design direction

This chapter outlines our project's scope, design goals, interaction vision, and sets specific criteria and testable targets. This framework guides our design efforts and provides measurable standards for evaluating progress and effectiveness, laying the groundwork for a user-centered design outcome.

Scope

Design goal and
interaction vision

Design criteria and
testable targets



Scope

To kickstart our redesign process, we've defined a clear scope for ourselves. This scope outlines the key elements that are central to our redesign efforts and will serve as the context for our work. Below, we've highlighted the key terms that will guide our design decisions and help us create a cohesive and effective redesign.

Regular Cooks

People that enjoy cooking and do it regularly

Home setting

The design context is within the users home

The product

The focus will only be on the product

Secondary function

Celcius Cooking
PLUS Cooking

Celcius Accessoires

Probe
Celcius Skillet

Onboarding

The part of the user journey will be the onboarding while cooking

Redesign scope

To narrow the scope of the redesign a distinction between the primary, extended and passive redesign focusses has been made (see figure 12). We made this distinction in order to effectively resolve our problem statement.

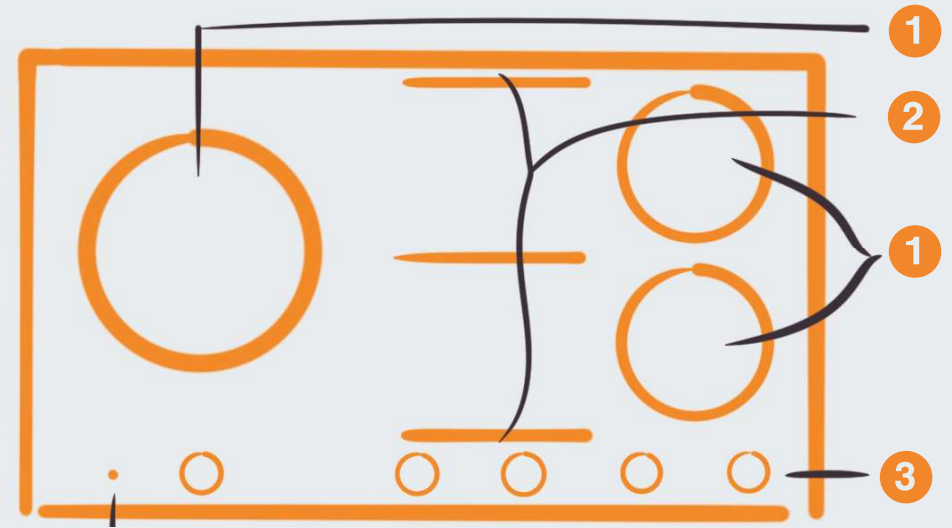


Figure 12: Overview redesign focusses

Primary redesign focus:

The elements below are prioritised for the redesign.

- 9 **Temperatures PLUS cooking**
There are only a few temperatures to choose from using the PLUS cooking, which are too hot and cannot be altered. this does not make the user feel in control.
- 13 **Lock PLUS cooking**
Whenever people set a PLUS cooking setting it is hard to get out of this setting. It either happens on accident or not at all. This is not intuitively
- 7 **The preheating fase**
The preheating fase stresses people out and they feel like they cannot make their own decision.
- 12 **No memory for Celsius settings**
Whenever you go out of the Celsius cooking, there is no memory to go back to the previous settings. It takes time to set it all up again.

Extended redesign focus:

The elements below are of lesser priority for the redesign.

- 5 **Wifi button**
The wifi button is confusing for a lot of users. They think it is a button for connecting the probe/skillet, but it is for the app. This is not intuitive.
- 4 **Pause button**
The pause buttons placement is not intuitively placed. It is used when users are stressing and feel stuck in using the plate.
- 14 **Placement PLUS and Celsius cooking**
The placement of the PLUS and celsius cooking function are not clear. The users often do not know in which functions they are.
- 6 **The letters for the time**
The letters for the time are not always clear. It is confusing that the letters are in English.
- 10 **Flickering temperatures Celsius**
When in Celcius cooking the iris shows the temperature you want it to be and the temperature the plobe measures. This flickers around a lot. The user feel confused and not in control.
- 11 **The settings**
If the user wants to change their settings, this does not work when there is a pan detected on the plobe. The user does not understand this intuitively.

Passive redesign focus:

The elements are important insights, but beyond our scope.

- 1 **Pan detection**
The pan detection is responding a bit slow. This makes the user confused for a small amount of time.
- 2 **Bridging**
The bridging function is not always intuitively and also responding a bit more slow. But after a while people figured the bridging out.
- 3 **Bluetooth connection**
The bluetooth connection of the plobe and skillet is not intuitive and does not always respond. This makes the users feel frustrated.
- 8 **Stirring and the plobe**
Whenever the plobe is working and you begin to stir the pan, the plobe has a malfunction and does not work anymore.
- The manual**
The manual is not clear on every page. Users struggled with finding what they were looking for.

Design goal and interaction vision

Design goal

We aim to redesign the Atag Induction Celsius Cooking Hob and equip **regular cooks** with **the confidence** to let the ATAG cooking plate **empower** them through their culinary experimentation in **an intuitive manner**. Our goal is to make them **feel in control** and **excited** about the cooking process, with the integration of new technology that Atag unlocks in their daily lives.

Interaction vision

We want the interaction with the redesign feel like you are playing the most beautiful improvised solo while jamming with your band. The user is the soloist and the band is the cooking plate.



Interaction qualities

Guiding

A well guided soloist by his band, is a soloist that feels confident and someone who can build upon their band.

In control

A soloist that is in control is an soloist that feels free to experiment with the music on their own and knows what the band will do.

Excited

A soloist that feels excited, is a soloist that is proud of his/her solo. He or she strived for more and succeeded.

Intuitive

A soloist that is intuitive, trust on his/her own skills and can take actions without thinking out how.

Confident

A soloist that feel confident knows what he or she can do and what the band will do. He or she is confident that the solo will be beautiful.

Design criteria and testable targets

Design Criteria

Testable targets

USER EXPERIENCE

- Users should feel **guided** in their cooking process.
- Users should feel a **boost of confidence** by **learning** new cooking skills/techniques with **ease and curiosity** instead of frustration.
- Users should be able to feel like they are **in control**.

USER INTERACTIONS

- The hob's **button placements** and **functionalities** should be **intuitive** to **navigate**.
- When users experience errors or interact with buttons/systems, **feedback** should **calm** them and help them **understand** what is happening.
- The hob should add **efficiency** and **ease** to the **cooking experience** rather than act as an obstacle to it.
- The hob should be something that acts as a user's **tool** to **improve** their **cooking skills**.

- **Moving on to the next step of cooking tasks** is **easy** and **understandable** for users and should get easier with second use.
- With the **PLUS** and **Celsius** functions, users should be able to **explore** without being scared of making mistakes or being locked into the functions.
- The **new buttons** (icons and placements) should be **intuitive** with a flatter learning curve than the current buttons.
- The **redesigned feedback** system should be **informative** while alerting the users in a calm manner.
- In depth **user tests**, accompanied by evaluative tools, would help gain insight into the hob's **usability**, the user's **experience** and help predict the **impact** on their cooking over time.

Design Criteria

- Users should feel **confident** in their **understanding** of the **hob's elements** (interface & accessories).
- To make it **easier to learn** and **use the hob** from the very **first time**, there should be **consistency** in the **usage** and the **feedback** from the interface.
- Functionalities should be **controllable/adjustable**, so **users are able to control the hob** instead of being controlled by it.
- **Celsius cooking** should be a functionality that **provides users with knowledge** about cooking with **degrees** while still allowing room for **self-exploration**.

Testable targets

- Users should be able to **understand** the features of the system **without having to consult the manual frequently**.
- If they do **consult** the **manual**, topics should be **easy to find** while being **written in a clear manner**. Consulting the manual **should be decreased with each time the user uses the hob**.
- Users should be able to **adjust the presets** and the **temperature** in **Celsius** cooking, while also **gaining knowledge** on precise temperature through the interface itself.
- In depth **user tests**, accompanied by evaluative tools, would help gain insight into the hob's **usability**, the user's **experience** and help predict the **impact** on their cooking over time.

Next steps

Plan for Phase 2

Now that Phase 1 is completed, our findings have led us to understand the scope for redesign on the hob's interface along with other factors in the overall user cooking experience. The details of our next steps towards the initial redesign proposal for Phase 2.

Brainstorm Ideas

A diverging phase that takes place after problem finding, where we freely explore all of our ideas and draw it out on one board.

Selection Round

A converging phase where we narrow down to top ideas and 2-4 concepts. Evaluation of the concepts against our findings also takes place here.

Concept

Select a single concept (From the ones above) that is in line with the design goal and interaction vision.

Initial Redesign Proposal

Create the first version of the redesign proposal which will be iterated on in the third phase.

References

Product Description

Atag Manual, <https://support.atag.nl/support/solutions/folders/77000323149>

Inductiekookplaten with Celsius°Cooking™, <https://www.atag.nl/collectie/kookplaten/celsius-kookplaten>

Apple Store, <https://apps.apple.com/nl/app/connectlife/id1519398796>

Usability Analysis

User Experience Questionnaire, <https://www.ueq-online.org>

Online Reviews

Coolblue, <https://www.coolblue.nl/en/product/933678/atag-bih09571ev.html>

Keukenloods, <https://www.keukenloods.nl/hi29571sv.html>

Consumentenbond, <https://www.consumentenbond.nl/inductiekookplaat/producten/atag/hi29571sv>

YouTube Videos, <https://www.youtube.com/watch?v=oROBNIPv1Gk>

Appendix

A: User flow

B: Onion of use

C: Survey first hand use

D: Cognitive walkthrough

E: Questionnaire

F: Moderator Script

G: Observer form

H: Data & analysis of usertest

I: Overall experience graphs

J: Overall emotion graph

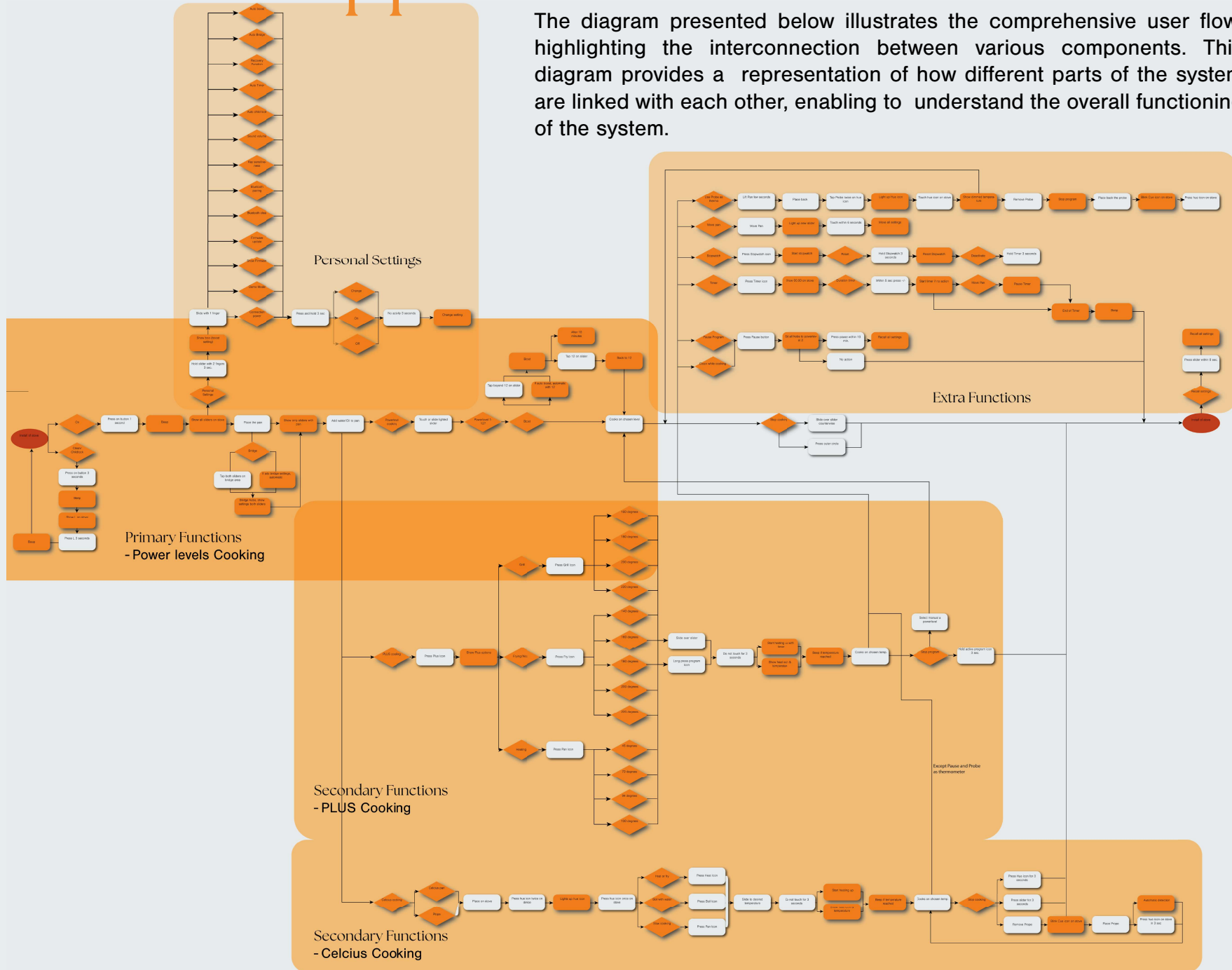
K: UEQ Questionnaire

L: Visuals UEQ Questionnaire



Appendix A: User flow

The diagram presented below illustrates the comprehensive user flow, highlighting the interconnection between various components. This diagram provides a representation of how different parts of the system are linked with each other, enabling to understand the overall functioning of the system.

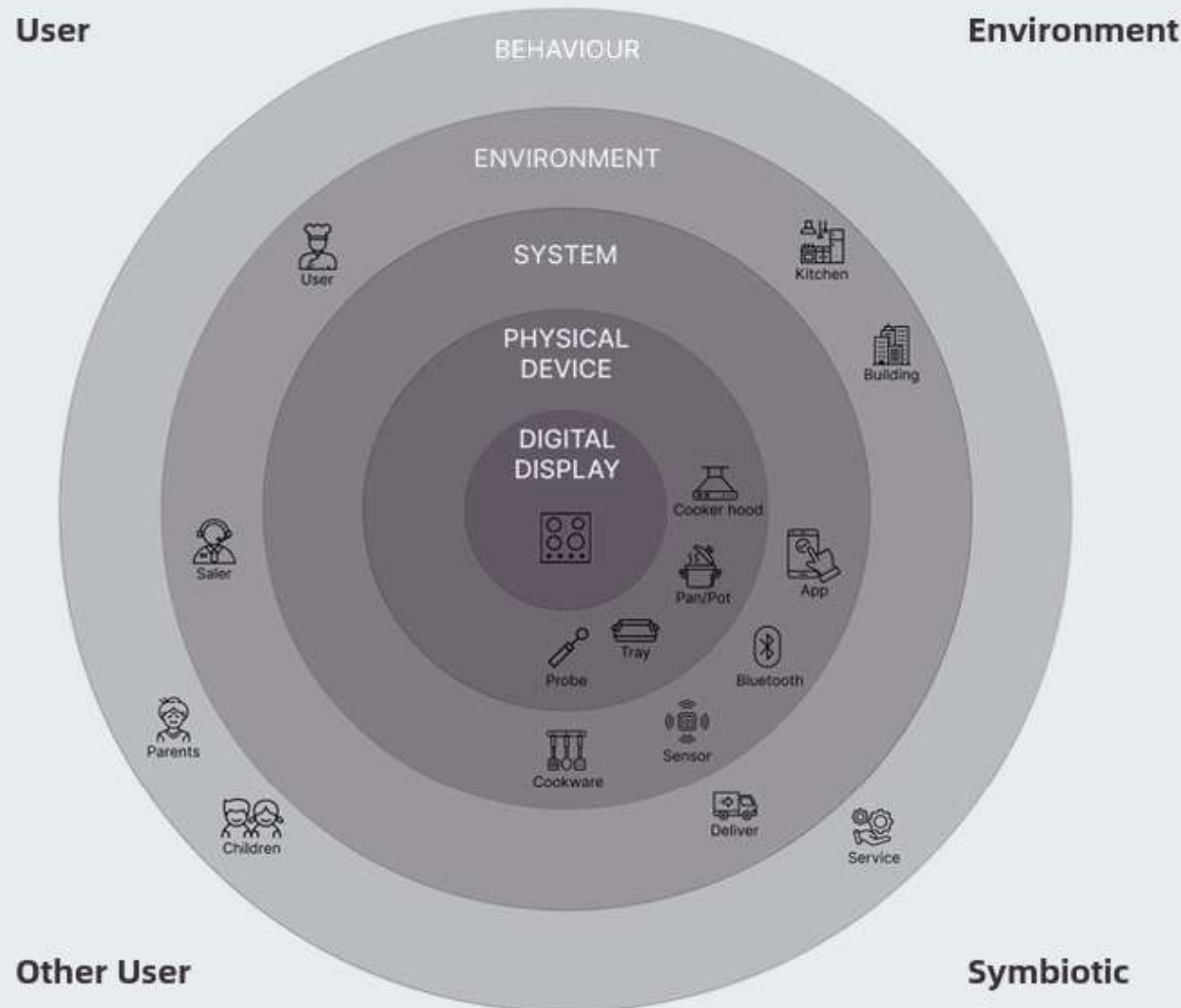


Appendix B: Onion of use

In the context of use for the ATAG temperature-controlled induction hob, we can explain from the following four perspectives:

This layer focuses on the person who directly interacts with the induction hob. For example, a home chef may need a device that can precisely control temperature for cooking meals, ensuring quality and taste. This requires the induction hob to have an intuitive interface, simple operation, and functionality that meets their cooking habits and needs.

User



The environmental context refers to the physical and social setting where the induction hob is placed and used. For instance, in a compact kitchen, the induction hob's size and shape should fit the space without disrupting the use of the kitchen. Additionally, how it integrates into modern kitchen design, as well as stable operation under various climate conditions (like temperature, humidity), are important considerations.

This layer is concerned with the product's interaction with other devices or systems. The ATAG induction hob might connect with other smart home devices in the kitchen, such as smart sockets, sensors, or an APP control system, forming an integrated smart kitchen ecosystem. For example, by connecting to the hob via an app for remote temperature control, or adjusting heat distribution during cooking through sensors, a more efficient and personalized cooking experience can be achieved.

At this level, consideration is given to those who might indirectly affect or be affected by the product, such as family members. Safety concerns for children, usability needs for the elderly, or the collective concern about energy consumption and efficiency are factors that need to be considered in the design.

Appendix C: Survey First-Hand use no manual

We wanted to capture our immediate response to the cooking hob without prior knowledge or reading the manual. We created a quick form and recorded our initial reactions to accomplish this. The graphs below depict our survey results, providing a proper first impression of the product.

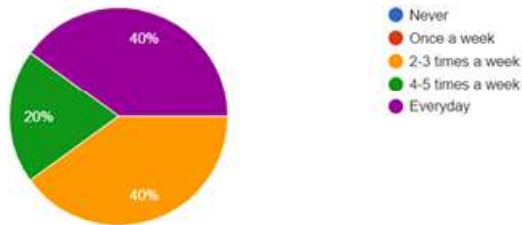


Appendix D: Cognitive walkthrough

The data for the cognitive walkthrough conducted with all five members of the group can be seen here. It started with noting down our perceived cooking abilities. Data including confidence, emotions and comments was then gathered step-by-step from basic functionalities to PLUS functions to the Celsius functions and lastly some extra functionalities.

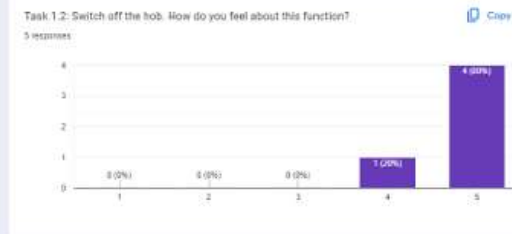
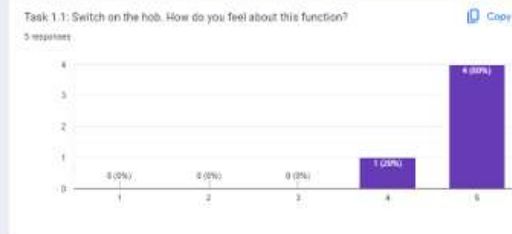
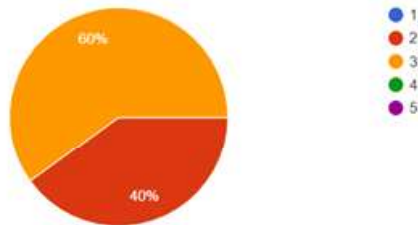
How often do you cook?

5 responses



When you are cooking, how many hobs do you use at the same time?

5 responses



Comments for Task 1

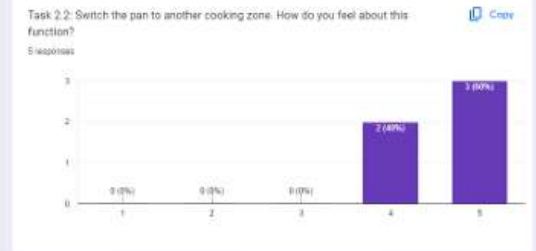
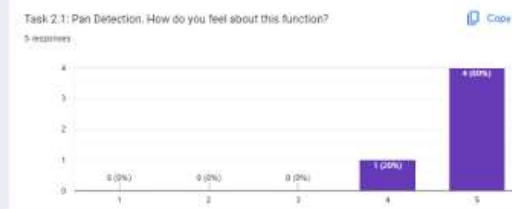
4 responses

Easy

You can also hop from 1 to 7 that is nice, but something I figured out later

very easy and satisfying sound

Easy and smooth. loved the unlocking



Comments for Task 2

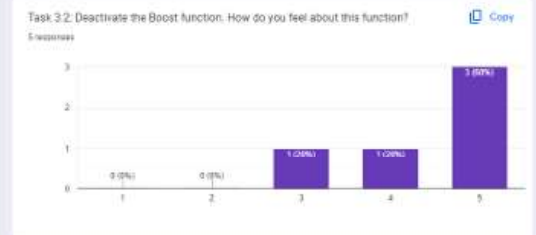
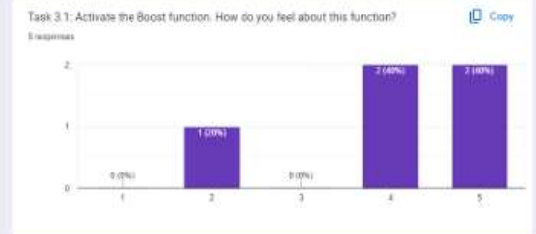
4 responses

It is sensitive

make sure to lift otherwise you will scratch

I like that you don't have to do anything, it takes some time tho to detect, so I was waiting if it reacted good

Love the clicking feeling of it too



Comments for Task 3

5 responses

Heat up very fast

You need to take care with turning it down, might need multiple tries

Why is this call the boost if it is only turning it on high? Confusing. I also really hate the sound

you have to know that you need to move clockwise and you need to know that you can turn it off by pressing

Easy but don't like the noise. it's too pitchy. Also may feel like u doing something wrong

Appendix D: Cognitive walkthrough

What do you think you can cook at the different levels (your assumptions)? (For example: 1-5 I feel like I can cook....)

5 responses

5

1-5 is sauteing, 6-10 is grilling, 10-12 is boiling

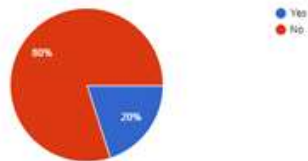
I just use the higher pitches when in a hurry or if I am not patient enough. For the boost, I also only use the lower once if your cooking a sauce that has to go slowly

1-5 slow cooking
6-9 sauteing
10-11 boiling
12 boost for fast boiling

I would usually cook slow up to 5 or 6 and use the rest for fast cooking. I see it as two to three ranges of cooking.

Did you use the manual for this section?

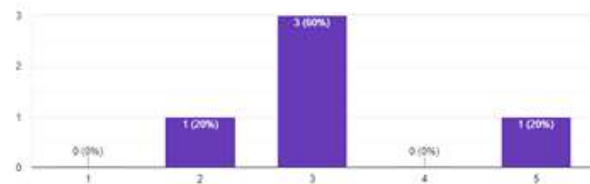
5 responses



Using the Cooking Hob - PLUS Cooking Mode

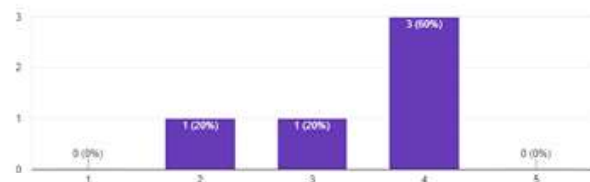
Task 4.1: How well do you understand this function?

5 responses



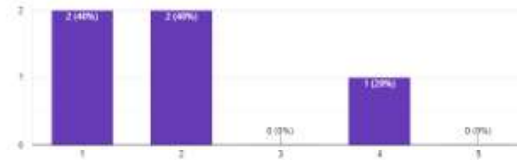
Task 4.2: How well do you understand this function?

5 responses



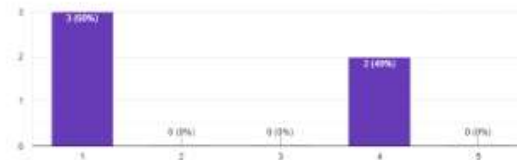
Task 4.3: How well do you understand this function?

5 responses



Task 4.4: Manually activate and deactivate the grill. How do you feel about this function?

5 responses



Comments for Task 4

4 responses

I could not find how to put on the grill function

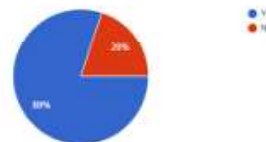
Stupid. Why does it go on a lock and how can I delete this lock or do I have to wait until it is heated. Icons are very clear to me! Manually grill function, it doesn't work!!

confused how to turn off the grill

It was alright The grill function can be made easier with automatic grill on.

Did you use the manual for this section?

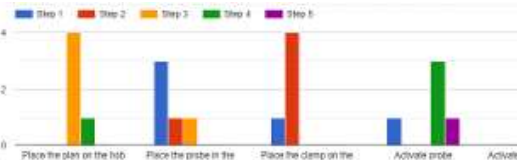
5 responses



Using the Cooking Hob - Hestan Cue Cooking Mode

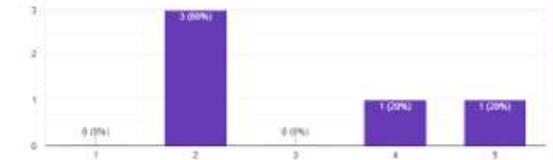
Task 5.1: What is the order of using this mode? First step:

4 responses



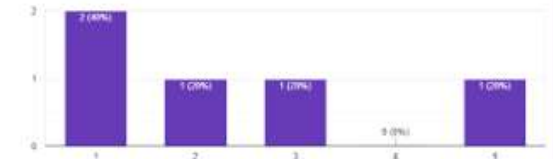
Task 5.2: Activate the probe. How do you feel about this function?

5 responses



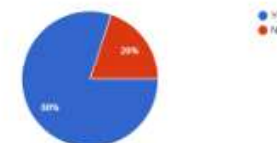
Task 5.3: Deactivate the probe. How do you feel about this function?

5 responses



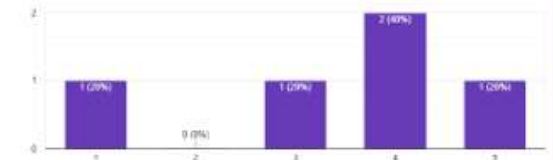
Task 5.4: Would you remove the probe while cooking?

5 responses



Task 5.5: Use the probe as a cooking thermometer. How do you feel about this function?

5 responses



Comments for Task 5

4 responses

Complex but sometimes useful

It is not clear to me how to activate and deactivate the probe. It does not show the arrow sign on the hob.

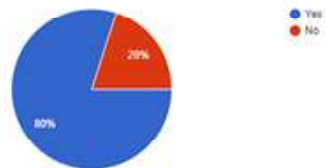
To many functions, it does not work for me, don't like it but do like the idea.

for more degrees you keep on going round and round on the slider, using the probe itself seems complicated and would get in the way of stirring for sure

Appendix D: Cognitive walkthrough

Did you use the manual for this section?

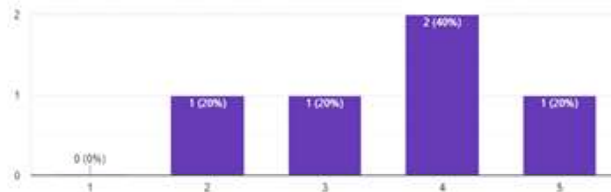
5 responses



Extra Functionalities

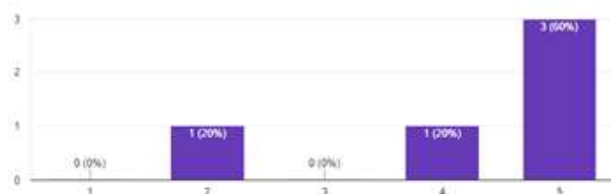
Task 6.1: Activate the personalisation 'Settings'. How do you feel about this function?

5 responses



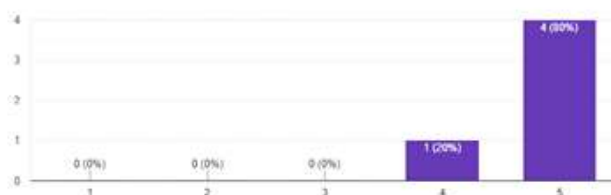
Task 6.2: Recall the settings after switching the appliance off. How do you feel about this function?

5 responses



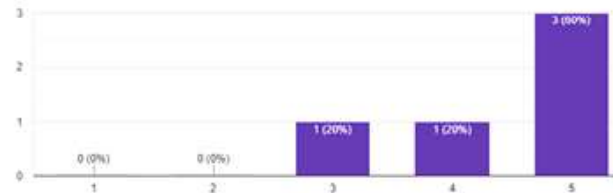
Task 6.3: Pause the cooking. How do you feel about this function?

5 responses



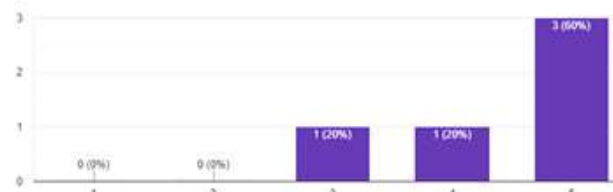
Task 6.4: Unpause the cooking. How do you feel about this function?

5 responses



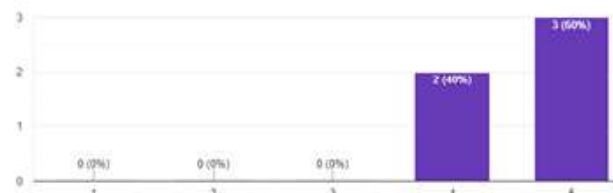
Task 6.5: Start and stop the timer function. How do you feel about this function?

5 responses



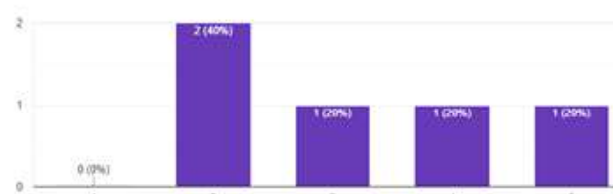
Task 6.6: Start and stop the stopwatch function. How do you feel about this function?

5 responses



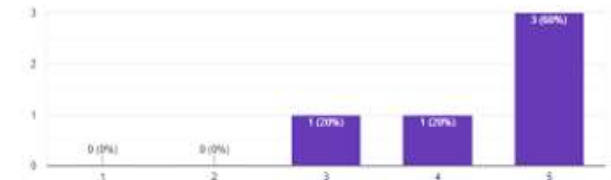
Task 6.7: Activate the WiFi. How do you feel about this function?

5 responses



Task 6.8: Activate the child lock for cleaning or for your hypothetical children. How do you feel about this function?

5 responses



Did you notice any audio or visual feedback like noises/beeps/symbols? What do you think about the responses or feedback from the appliance?

5 responses

The feedback beeps are the same when I choosing different functions

Yes there is a sound for most settings, but to me it is not clear that the timer is set

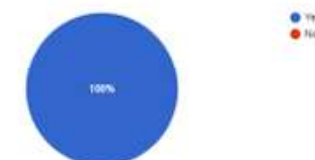
I like the noise of the timer on stopwatch. I don't like the other harder beeps, but maybe it is getting better when you are cooking

I like the beeps clear sharp but not annoying

enjoyed the beeps and noises. they are calm yet loud enough to alert the user. however sometimes the cooking noises can make u feel like ur doing something wrong

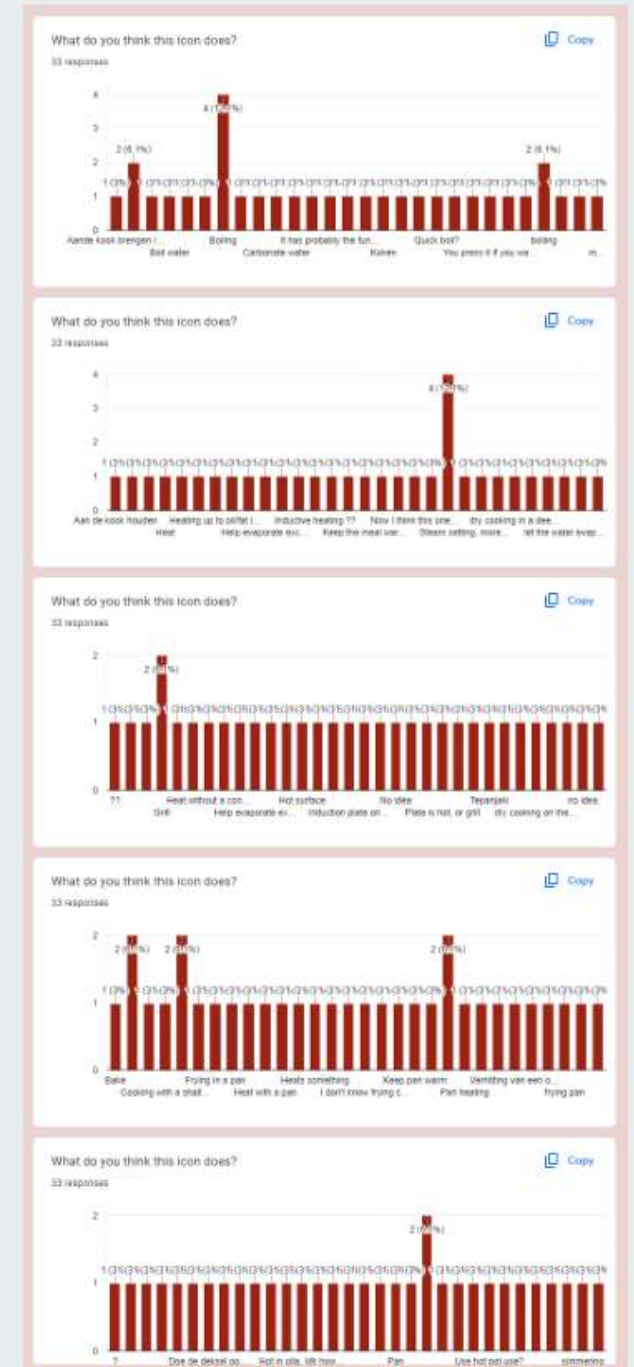
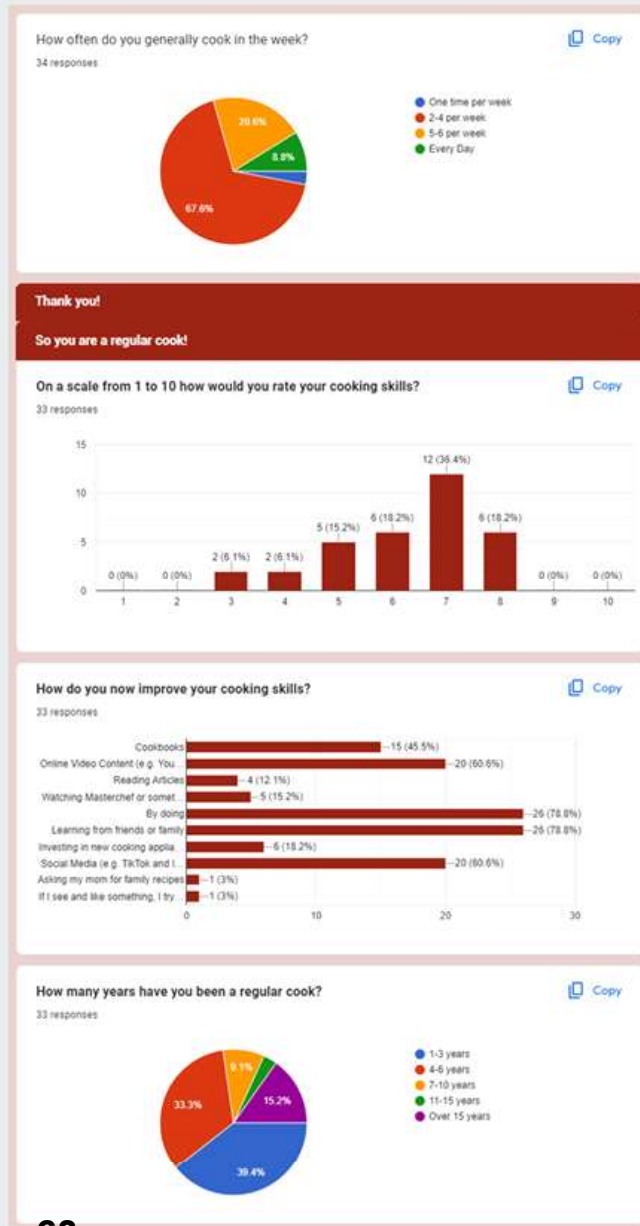
Did you use the manual for this section?

5 responses

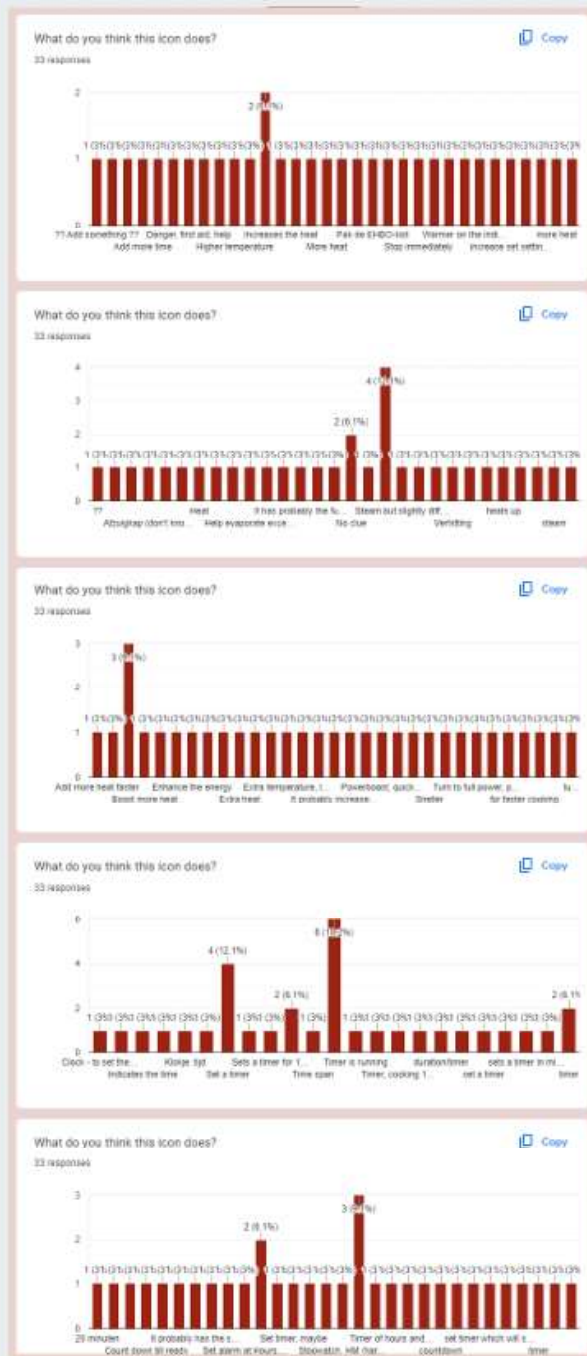


Appendix E: Questionnaire

Here is all the data gathered from 33 participants in the form of a survey.



Appendix E: Questionnaire



Plus Cooking settings

33 responses

1

Heat a frying pan

keeping it hot

broil perfectly

Frying in a pan as opposed to a deep fryer

for pan

let water evaporate from pressure cooker

Cooking with a fry pan on high, like pancakes or chicken filet

Heat endured by a pan

Plus Cooking settings

33 responses

2

Heat a grill

keeping it hot, but dont really know what the difference is from this icon and the previous, outside that the previous is a pan but I dont know what this icon stands for

no clue

For special appliance that can be used on a stove

for something without handle?

dehydrate something that is in the cooker

General, on high

Heat with a plate

Hue Cooking

Within the Hue cooking settings, what cooking zone do you think this icon implies?

33 responses

1

Heat water

lukewarmish

non boiling heating

Warming up soup

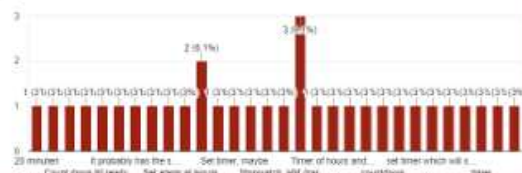
you can put the pot here

Keep a constant temperature

slowly heating up a liquid from cold.

Heat below 100 Celsius degree

Appendix E: Questionnaire



Heat till the liquid is boiled 100 Celsius degree

Reducing liquid

Try the big circle?

Try to press the big circle

Question	Percentage
1. I would tap the "pi..."	1.00%
2. Maybe -	1.00%
3. No idea	1.00%
4. Press on the -	1.00%
5. Press small icon u...	1.00%
6. Press the sections...	1.00%
7. Same answer	1.00%
8. Touch "minus"	1.00%
9. long press in the c...	1.00%
10. press on...	1.00%
11. ...	1.00%
12. ...	1.00%
13. ...	1.00%
14. ...	1.00%
15. ...	1.00%
16. ...	1.00%
17. ...	1.00%
18. press on the	2.00%
19. ...	1.00%
20. ...	1.00%

aayushi31@gmail.com

Appendix F: Moderator Script

Moderator Script for ATAG Celsius Cooking Hub Usability Test

Introduction (2 minutes)

Greeting

- Thank you for participating in our usability test of Atag Celsius Induction hob today. Your feedback is very important in improving the user experience of this cooking hob. The session will take about an hour ending with a delicious meal, and we'll go through a series of tasks ranging from simple to more advanced functionalities. Feel free to share all your thoughts and reactions as we proceed.

Consent form (2 minutes)

- Filled by participant before the test

Questionnaire (5 minutes)

- How often do you generally cook in the week?
- What do you do first before using a new appliance?
- How familiar are you with induction cook?

Priming (6 minutes)

- And I would like to ask some questions
 - At what time of the day do you usually cook?
 - If you would have this cooking plate new, what would you do/what steps would you take to get familiar with the product? (
 - Follow up questions based on answer (for example, if they say they use the manual, what part of it allow them to take you through their real-life scenario)
- The Celsius Cooking Hub offers a variety of functions designed to enhance your cooking experience, including adjustable power levels, a timer, Celsius cooking, and a Plus cooking program. Today, you'll get to explore these features firsthand. [Not SHOW the functions but TELL them it exists (can show in the manual but not on the hob)]
 - Timer
 - Normal (power levels)
 - Plus (temperature and different cooking styles)
 - boiling, grilling, wokking
 - Celsius
 - bluetooth by tapping twice on the hob

- Explain how they should try thinking out loud during the cooking, say the steps they are taking out loud and what they think out loud.
 - How the appliance is being tested and not them
 - Need to let them use it as they do in real life
 - The manual is available to use at anytime throughout

Starting the Test:

TIPS:

- Stand by and just observe what participant does. Try not to step in the testing process.
- Clean the plate in between really well (clean off the fingerprints)

SCENARIOS

- Simple task (5 mins) - Basic functions
 - Switching the hob on and placing the pan
 - SCENARIO: STARTING UP: Imagine you are starting to cook for the day at _____. Today you're going to make a homemade dahl and rice for yourself. You did a great job by prepping for this meal, and all you have to do now is cook. It is time to turn on the cooking plate and make it ready to use.
- Simple task (10 mins) - Primary functions
 - Boil rice
 - Set timer
 - Set up the water for boiling
 - SCENARIO: PREPARE THE RICE: Now the cooking plate is ready to use, it is time to cook the rice. Boil the water and set a timer.
- Let them fill out the UEQ while waiting
- Medium task (15 mins) - Cooking with PLUS
 - Onions, ginger, garlic, herbs
 - setting it to fry/wok setting
 - see the preheat function (wait for the timer to go down)
 - wait for the onion to add the vegetable
 - SCENARIO: STARTING THE DAHL: Great, the rice is totally ready to eat! Now only the Dahl, so let's start sauteing (fruiten) the onions. Luckily the hob has a Plus Cooking setting for this, that you could use!
 - Let them fill out the UEQ + ask comments/notes/thoughts
 - What did you think about the PLUS functions? Do you see yourself using it?
 - If yes, for what kind of dishes?
 - If not, do you see yourself learning it?

Appendix F: Moderator Script

- Expert task (25 mins) - Cooking with Celsius
 - Make Dahl
 - know which temperature you need to set (85-95°C)
 - set up the Hue settings
 - set timer
 - wait for the heat up
 - SCENARIO: FINISHING THE DAHL: The Dahl is almost ready. Time to add the last ingredients. This has to simmer on 90 degrees celsius precisely.
 - Let them fill out the UEQ + ask comments/notes/thoughts
 - What did you think about the Celsius functions? Do you see yourself using it?
 - If yes, for what kind of dishes?
 - If not, do you see yourself learning it?
 - Concluding test (10 mins) - Cleaning
 - Child lock
 - Wait until cold
 - Change settings
 - Let them fill out the UEQ
 - SCENARIO: WRAPPING UP: After having cooked, you want to clean your stovetop. Find yourself a way of cleaning the surface.
 - SCENARIO: You want to make boost cooking more accessible. Therefore, you want the hob to turn on boost automatically when set to power level 12. Try to change the personal settings to automatic boost.
 - SCENARIO: Show them the online Celsius recipes and ask their opinion.
-
- Session Closure: Confirm that all participant questions have been addressed, provide any agreed-upon compensation, and thank the participant again for their valuable input.
 - Guidance Out: Ensure the participant knows how to exit the building, and express appreciation for their time once more.
-

Appendix G: Observer form

Observer Form Template

Participant: _____

Date: _____

Observer name: _____

Primings

Objective: Understanding user

Observations: _____

What time of the day do you usually cook?

If you would buy a new cooking hob, what steps would you take to get familiar with the product?

Further questions and answers (if any)

Comments on the explanation of the product (if any)

Task 1: simple task - basic functions

Objective: turn on the hob and place the pan

Time taken: _____

Nr of steps: _____

Look for: turning on, pan detection

Observations: _____

Interaction flow different steps the user takes

Hesitations/confusions: looking at hob, waiting, vocal cues

Participant feedback when asked questions

Questions:

- What did you think about the functions?
- Additional thoughts

Body language facial expression, whole body

Appendix G: Observer form

Task 2: simple task - primary functions

Objective: boil rice, set timer and set up the water for boiling

Time taken: _____ **Nr of steps:** _____

Look for: turning motion heating, setting timer vs. stopwatch

Observations: _____

Interaction flow different steps

Hesitations/confusions looking at hob, waiting, vocal clues

Participant feedback when asked questions

Questions:

- What did you think about the functions?
- Additional thoughts

Body language facial expression, whole body

Task 3: simple task - cooking with PLUS

Objective: Cook onion and a vegetable using the fry/wook setting (preheat function)

Time taken: _____ **Nr of steps:** _____

Look for: finding PLUS mode, preheat countdown, switching modes

Observations: _____

Interaction flow different steps

Hesitations/confusions looking at hob, waiting, vocal clues

Participant feedback when asked questions

Questions:

- What did you think about the functions? Do you see yourself using it?
 - If yes, for what dishes?
 - If no, would you learn?
- Additional thoughts

Body language facial expression, whole body

Appendix G: Observer form

Task 4: simple task - cooking with CELSIUS

Objective: Make Dahl, set temperature, and use Hue settings

Time taken: _____ **Nr of steps:** _____

Look for: temperature setting, switching temperatures

Observations: _____

Interaction flow different steps

Hesitations/confusions looking at hob, waiting, vocal clues

Participant feedback when asked questions

Questions:

- What did you think about the functions? Do you see yourself using it?
 - If yes, for what dishes?
 - If no, would you learn?
- Additional thoughts

Body language facial expression, whole body

Task 5: small concluding task - cleaning

Objective: Activate child lock, adjust settings for cleaning

Time taken: _____ **Nr of steps:** _____

Look for: setting child lock, adjusting settings

Observations: _____

Interaction flow different steps

Hesitations/confusions looking at hob, waiting, vocal clues

Participant feedback when asked questions

Questions:

- How do you feel about cleaning up?
- How do you feel about the settings?
- Opinion of recipes
- Additional thoughts

Body language facial expression, whole body

Appendix G: Observer form

Overall experience

Questions:

- What is your overall experience of the hob?
- Would you buy this? Why or why not?

General feedback:

Intuitive aspects

Confusing aspects

Satisfaction level

Improvement suggestions

Additional notes

Appendix H: Data & analysis of user tests

About the participants



Participant 1

Expertise level: 8 out of 10

Type of cook: Participant 1 is in a cooking club, here he likes to experiment with different new function and new recipes. He does not follow the recipes strictly

Times of cooking per week: 3-4 times a week

First steps for using a new cooking hob: He would first start up slowly and start figuring out by him self. When ever he is stuck he will look at the manual.



Participant 2

Expertise level: 7 out of 10

Type of cook: likes to cook for guests and test out new recipes but does not always have the time to spend a lot of time in the kitchen.

Times of cooking per week: 5-6 times a week

First steps for using a new cooking hob: start playing around with it, without reading the manual.



Participant 3

Expertise level: 6 out of 10

Type of cook: likes to try new recipes, cooks a lot of Indian food.

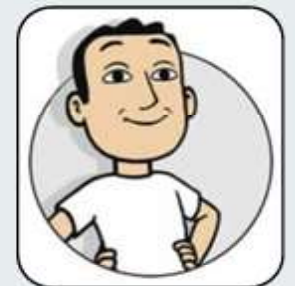
Times of cooking per week: 5 - 6 times a week

First steps for using a new cooking hob: would first read the manual, but does not like to read a lot.

Appendix H: Data & analysis of user tests

Participant 1 Task 1: Simple Task - Using Basic Functions

Priming		Task 1: simple task - basic functions		
Objective: What does the user have to do?	Step 1.1 understand the user	Step 1.1 turn on the plate	Step 1.2 Seeing pan detection/ symbols	Step 1.3 turn on the hob
Observations: what time of the day do you usually cook?	5 - 6 pm after work			
Observations: if you would buy a new cooking hob, what steps would you take to get familiar with the product?	connect and start using see manual only when start			
Observations: Further questions and answers	he is already familiar with the hob uses to heat food with basic functions Not really uses the hob for anything 3 times a week			
Effectiveness: Was the user able to complete the task?				
Errors: What difficulties did they encounter?			Did not understand the symbols Did not see the symbols Did not see the symbols	There was a small delay in the pan detection
Satisfaction: How satisfied did they feel with the task and product in that task?		Yes/No Why? 1 minute	Yes/No Why? ... minutes and ... seconds	Yes/No Why? ... minutes and ... seconds
Efficiency: How long did it take to complete the task?				
Experience: What was the overall experience with the product during this task?				
Quotes: What did the participant say out loud during this task that is of importance?				
Other: Are there any other things that stood out during this task that are of importance?				



Appendix H: Data & analysis of user tests

Participant 1 Task 2: Simple Task - Cooking with Primary Functions (Normal Mode)

Task 2: simple task - primary functions		Experience: What was the overall experience with the product during this task?	Sentiment Analysis	
Objective: What does the user do? Write down every step	Step 2.1 Finding and using the primary setting		Step 2.2 Using the timer	positive neutral negative
Effectiveness: Was the user able to complete the task?	easy medium hard			
Errors: What difficulties did they encounter?	First he went to the plus/minus functions Confused about the temperature notation Not quite sure if it is working	Second, different plus and power, plus without nothing Confused about why the temperature for cooking He was unsure of the effectiveness because of the plus and minus	Confused about the timer notation He was unsure of the effectiveness because of the plus and minus	45 celsius or 4.5? Huh? I think it is going on now I didn't understand clocks, but now I do It's clear it worked, it's fast Scratching this, figures it out There is no memory
Satisfaction: How satisfied did they feel with the task and product in that task?	Yes/No Why? 8 minutes and ... seconds	Yes/No Why? ... minutes and ... seconds		
Efficiency: How long did it take to complete the task?				
		Other: Are there any other things that stood out during this task that are of importance?		



Appendix H: Data & analysis of user tests

Participant 1 Task 3: Medium Task - Cooking with PLUS

Objective: What does the user do? Write down every step	Step 3.1 Turning on the plus cooking settings	Experience: What was the overall experience with the product during this task?	positive	
Effectiveness: Was the user able to complete the task?	easy		neutral	●
	medium		negative	
Errors: What difficulties did they encounter?	hard	Quotes: What did the participant say out loud during this task that is of importance?	Not very clear to me	How do I adjust it
	Very confused by approach, unsure the only way I managed to cook		Not a reason to buy	very quickly, already coloring
	the pan got very hot in the frying function		I like how you've made the controls (touchscreen)	Not for me, but maybe for other people
	got stuck with one kind of plus cooking, tried but after some time they stopped		When you're using me, I don't want to miss about having to do it for me	I would definitely skip this function
	Things to fix: the right temperature, having the controls a bit more		When I would have the product, I would be using it more often	I like inventing new functions
Satisfaction: How satisfied did they feel with the task and product in that task?	Yes/No	Other: Are there any other things that stood out during this task that are of importance?		
	Why?			
				
Efficiency: How long did it take to complete the task?	5 minutes and ... seconds			



Participant 1 Task 4: Hard Task - Cooking with Celsius

Participant 1 Task 4: Hard Task - Cooking with Celsius



Appendix H: Data & analysis of user tests

Participant 1 Task 5: Small Concluding Tasks

	Step 5.1 Cleaning	Step 5.2 Settings	
Objective: What does the user do? Write down every step			Experience: What was the overall experience with the product during this task?
Effectiveness: Was the user able to complete the task?			
Errors: What difficulties did they encounter?	- he does not use the lock function and did not know - a button is accidentally pressed - turns of cooking plate after cleaning - he does not use the lock function and did not know	- goes to manual to see how it works - first only uses one finger instead of two	- I don't use the lock, but it is not really a need, just a button that is not really clear - I don't know why the light is changing, but the function is not - match that clear, there are too many shelves
Satisfaction: How satisfied did they feel with the task and product in that task?	Yes/No Why? It is just very simple to use, I don't use the function of the cooking plate	Yes/No Why? He thought it was fine, only a lot to read	Quotes: What did the participant say out loud during this task that is of importance?
Efficiency: How long did it take to complete the task?	4 seconds	2 minutes	Other: Are there any other things that stood out during this task that are of importance?

Appendix H: Data & analysis of user tests

Participant 1 Overall Experience

Satisfaction: How satisfied did they feel with the task and product in that task?

he liked
it very
much

looking
forward in
tasting the
food

excited
about the
new
function

maybe get it
in the future,
but maybe
second hand

**Improvement
suggestions:** What does
the participant think can be
improved about the current
product

if atag want to
do everything,
people get lost

sometimes
it is best to
have less

Participant 2 Task 1: Simple Task - Using Basic Functions

Participant 2 Task 1: Simple Task - Using Basic Functions

Appendix H: Data & analysis of user tests

Participant 2 Task 2: Simple Task - Cooking with Primary Functions (Normal Mode)

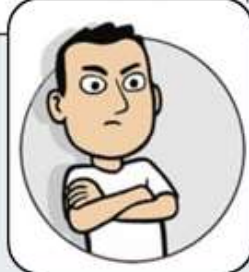
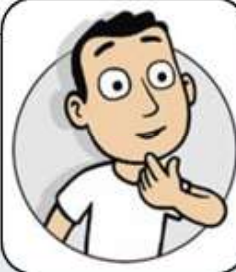
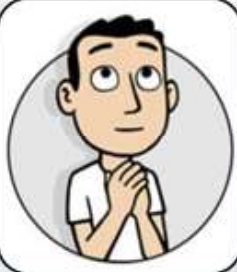
	Step 2.1 put rice in pot	Step 2.2 put temperature slider	Step 2.3 put timer
Objective: What does the user do? Write down every step.			
Effectiveness: Was the user able to complete the task?			
Errors: What difficulties did they encounter?			   
Satisfaction: How satisfied did they feel with the task and product in that task?	Yes/No Why? 	Yes/No Why? 	Yes/No Why? 
Efficiency: How long did it take to complete the task?	2 minutes and 15 seconds		

Experience: What was the overall experience with the product during this task?


Quotes: What did the participant say out loud during this task that is of importance?



Other: Are there any other things that stood out during this task that are of importance?

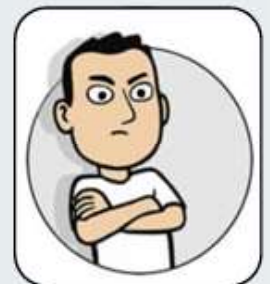


Appendix H: Data & analysis of user tests

Participant 2 Task 3: Medium Task - Cooking with PLUS

	Step 3.1 put the pan & turn on the normal function	Step 3.2 trying to find function & checking the manual for PLUS	Step 3.3 shifting pan to bridge and then bringing it back	Step 3.4 found the PLUS, adjusting buttons	Step 3.5 trying function (frying onions and ginger garlic)
Objective: What does the user do? Write down every step					
Effectiveness: Was the user able to complete the task?	easy				
Errors: What difficulties did they encounter?		could not find the function could not find the manual	could not find the function could not find the manual	could not find the function could not find the manual	could not find the function could not find the manual
Satisfaction: How satisfied are they with the task and product in that task?	Yes/No Why?	Yes/No Why?	Yes/No Why?	Yes/No Why?	Yes/No Why?
Efficiency: How long did it take to complete the task?	10 minutes				

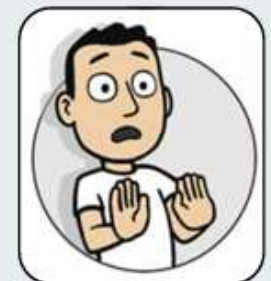
	Step 3.1 put the pan & turn on the normal function	Step 3.2 trying to find function & checking the manual for PLUS	Step 3.3 shifting pan to bridge and then bringing it back	Step 3.4 found the PLUS, adjusting buttons	Step 3.5 trying function (frying onions and ginger garlic)
Experience: What was the overall experience with the product during this task?	positive				
Quotes: What did the participant say and find during this task that is of importance?					
Other: Are there any other things that stood out during this task that are of importance?					



Appendix H: Data & analysis of user tests

Participant 2 Task 4: Hard Task - Cooking with Celsius

Objective: What does the user do? Write down every step	Step 4.1 put seasoning, water and start cooking	Step 4.2 trying to turn it on and put probe on the pan	Step 4.3 reading the manual to understand probe	Step 4.4 connecting probe/pan to hub	Step 4.5 Seasoning Dish!
Effectiveness: Was the user able to complete the task?	easy				
Errors: What difficulties did they encounter?		- couldn't find the correct amount of seasoning - couldn't find the correct amount of water - couldn't find the correct amount of probe	couldn't find the correct amount of probe	couldn't find the correct amount of probe	couldn't find the correct amount of probe
Satisfaction: How satisfied did they feel with the task and product in that task?	Yes/No Why?	Yes/No Why?	Yes/No Why?	Yes/No Why?	Yes/No Why?
Efficiency: How long did it take to complete the task?	13 minutes				



Appendix H: Data & analysis of user tests

Participant 2 Task 5: Small Concluding Tasks

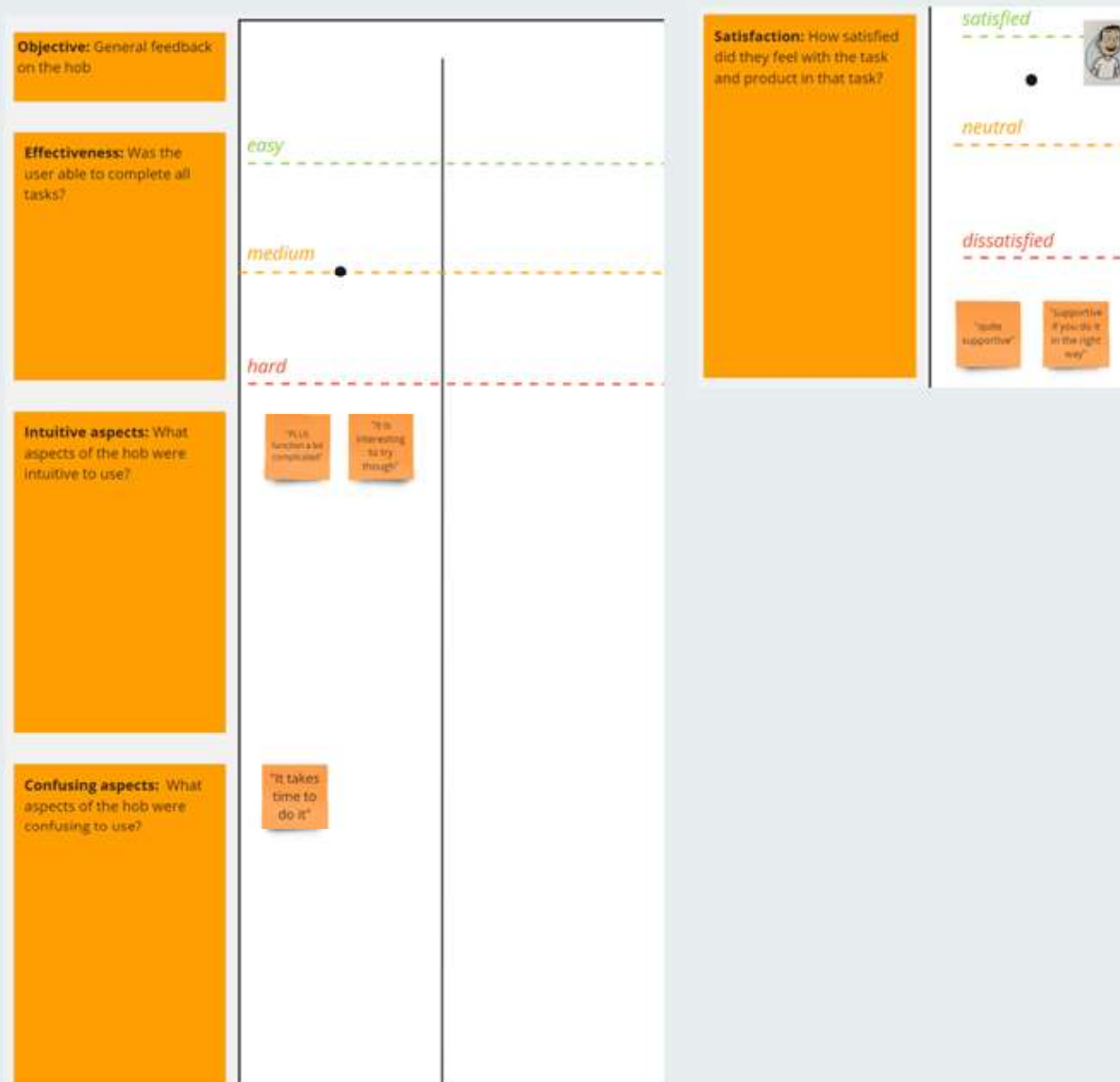
	Step 5.1 Takes the pans away from the hob	Step 5.2 Cleans hob (without locking)	Step 5.3 Settings
Objective: What does the user do? Write down every step			
Effectiveness: Was the user able to complete the task?	easy		
Errors: What difficulties did they encounter?			It did not work because the hob was locked and we didn't know the locking code. While cleaning the hob, it felt like it was not working properly. The hob is noisy when it is on and it is not clear if it is on or off. Read the manual but it was not clear if it was on or off. Got a lot frustrated when it didn't work easily.
Satisfaction: How satisfied did they feel with the task and product in that task?	Yes/No Why? 	Yes/No Why? 	
Efficiency: How long did it take to complete the task?	2 minutes		

	Step 5.1 Takes the pans away from the hob	Step 5.2 Cleans hob (without locking)	Step 5.3 Settings
Experience: What was the overall experience with the product during this task?	positive	neutral	negative
Quotes: What did the participant say out loud during this task that is of importance?	"The hob is noisy when it is on and it is not clear if it is on or off." "It's looking suspicious! How would I help it clean?" "I don't know cleaning without the manual." "I don't know how to lock it!" "I think it is on, not sure" "I don't see if it is on or off"	"I don't know cleaning without the manual." "It wasn't clear for me to lock it!" "I don't know how to lock it!" "I don't see if it is on or off"	"I don't see if it is on or off" "I don't see if it is on or off" "I don't see if it is on or off"
Other: Are there any other things that stood out during this task that are of importance?			



Appendix H: Data & analysis of user tests

Participant 2 Overall Experience



Appendix H: Data & analysis of user tests

Participant 3 Task 1: Simple Task - Using Basic Functions

Priming	
Objective: What does the user have to do?	Step 1.1 understand the user.
Observations: what time of the day do you usually cook?	every day after the turn 10:00
Observations: If you would buy a new cooking hob, what steps would you take to get familiar with the product?	read manual
Observations: Further questions and answers	looks a bit of a pain
Comments: Do they have any comments on the explanation of the product?	"oh okay"

Task 1: simple task - basic functions	
Objective: What does the user have to do?	Step 1.1 turn on the hob
Effectiveness: Was the user able to complete the task?	Step 1.2 Place the pan
Errors: What difficulties did they encounter?	
Satisfaction: How satisfied did they feel with the task and product in that task?	
Efficiency: How long did it take to complete the task?	

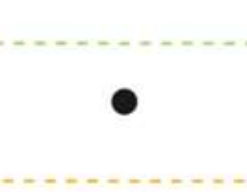
Step 1.1 turn on the hob	Step 1.2 Place the pan
easy	
medium	
hard	
wants to tap twice	thought when the pan is heating on the hob
he was fast and quite happy	he was fast and quite happy
0 minutes and 31 seconds	0 minutes and 47 seconds

Experiences: What was the overall experience with the product during this task?	Quotes: What did the participant say out loud during this task that is of importance?	Other: Are there any other things that stood out during this task that are of importance?

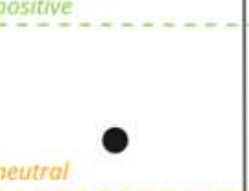


Appendix H: Data & analysis of user tests

Participant 3 Task 2: Simple Task - Cooking with Primary Functions (Normal Mode)

	Step 2.1 Uses PLUS settings to heat up pan	Step 2.2 Uses power levels to heat up pan
Objective: What does the user do? Write down every step		
Effectiveness: Was the user able to complete the task?		
Errors: What difficulties did they encounter?	he can't figure out if the pan is heating or not he can't figure out if the pan is heating or not used PLUS settings and therefore can't change the settings he can't figure out if the pan is heating or not he can't figure out if the pan is heating or not	he can't figure out if the pan is heating or not he can't figure out if the pan is heating or not he can't figure out if the pan is heating or not he can't figure out if the pan is heating or not he can't figure out if the pan is heating or not
Satisfaction: How satisfied did they feel with the task and product in that task?	Yes/No, Why?  he felt confident in use.	Yes/No, Why?  he felt confident in use.
Efficiency: How long did it take to complete the task?	2 minutes and 51 seconds	1 min minute and 43 seconds

Experience: What was the overall experience with the product during this task?



Quotes: What did the participant say out loud during this task that is of importance?

"I don't know what is happening"

"Is it on or off?"

"The hob gives the same noise when it does not heat up"

"Now I can't change it"

"I don't know how long cooking is going to take"

"Oh the other hob is still on"

"Is it hours or minutes"

Other: Are there any other things that stood out during this task that are of importance?

"We felt proud of himself" since he figured out how it worked

The hob was on child lock

He thinks he will be more familiar next time

He used + settings instead of power levels

he can't figure out if the pan is heating or not

The hob was on child lock

When the hob that there is a timer and stopwatch

he can't figure out if the pan is heating or not



Appendix H: Data & analysis of user tests

Participant 3 Task 3: Medium Task - Cooking with PLUS

Objective: What does the user do? Write down every step	Step 3.1 setting up PLUS cooking Step 3.2 turning off PLUS mode and using power levels		Experience: What was the overall experience with the product during this task?	positive neutral negative	
Effectiveness: Was the user able to complete the task?	easy medium hard				
Errors: What difficulties did they encounter?	could not reduce heat, but turned it off. This happened 3 times, gave over 5 minutes realised this is how that, but he can't adjust it why? what shall 140 degrees could not make more time, there is a countdown timer for the hob, heating up noticed the hob is more powerful than expected, so he had to use the heat		Quotes: What did the participant say out loud during this task that is of importance?	"Oh it started already" (on heating up) "I don't know how to adjust" "Do I use the dial to adjust?" "On tapping works better than dialing" "Is 140 the lowest?" "This is too much" (on the PLUS mode for causing errors) When he wanted to adjust the heat, before 140 is built in, got annoyed by that He could see it, but when he saw more explanation about the PLUS mode Once I got the explanation he understood it The hob was on child lock When did not "undo" in the same in PLUS settings as power levels After use explanation about PLUS settings	
Satisfaction: How satisfied did they feel with the task and product in that task?	Yes/No, Why? the hob does not do what he wants this works how he wants it		Other: Are there any other things that stood out during this task that are of importance?	The hob was on child lock	

Participant 3 Task 4: Hard Task - Cooking with Celsius



Participant 3 Task 5: Small Concluding Tasks

Participant 3 Task 5: Small Concluding Tasks

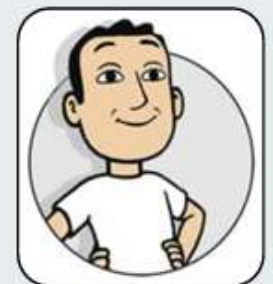
	positive	neutral	negative
Experience: What was the overall experience with the product during this task?			
Quotes: What did the participant say out loud during this task that is of importance?			
Other: Are there any other things that stood out during this task that are of importance?			



Appendix H: Data & analysis of user tests

Participant 3 Overall Experience

Objective: General feedback on the hob	Task 1.1 overall experience	Satisfaction: How satisfied did they feel with the task and product in that task?	<i>satisfied</i>
Effectiveness: Was the user able to complete all tasks?	<i>easy</i>		<i>neutral</i>
	<i>medium</i>		<i>dissatisfied</i>
Intuitive aspects: What aspects of the hob were intuitive to use?	looks nice easy to clean cools down fast	Improvement suggestions: What does the participant think can be improved about the current product?	would use smoother levels most heating is quite long time taking with the hob & grill the probe connection does not work now connecting the pan with hob/grill was not working either
Confusing aspects: What aspects of the hob were confusing to use?	the probe button is quite the same as the one for the hob, so it's a bit confusing the probe button is quite the same as the one for the hob, so it's a bit confusing the probe button is quite the same as the one for the hob, so it's a bit confusing	Additional notes: Any additional notes the participant gives on the overall experience	the overall emotion was it was trying new things works well if you get used to it

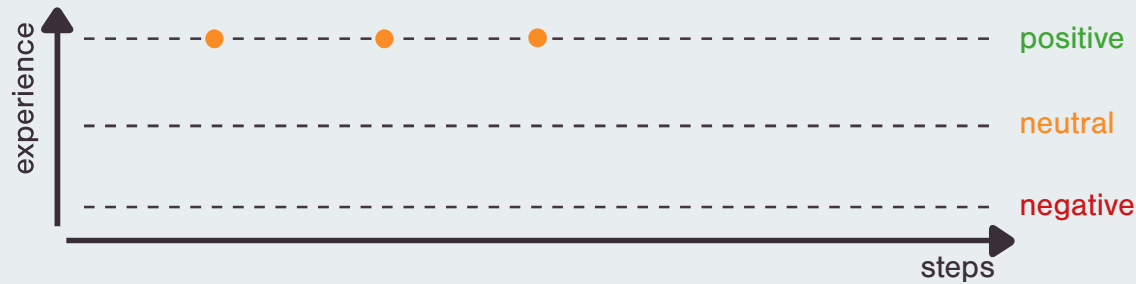


Appendix I: Overall experience graphs

Task 1: basic functions

Participant 1

Main pain points



Participant 2



Participant 3



Appendix I: Overall experience graphs

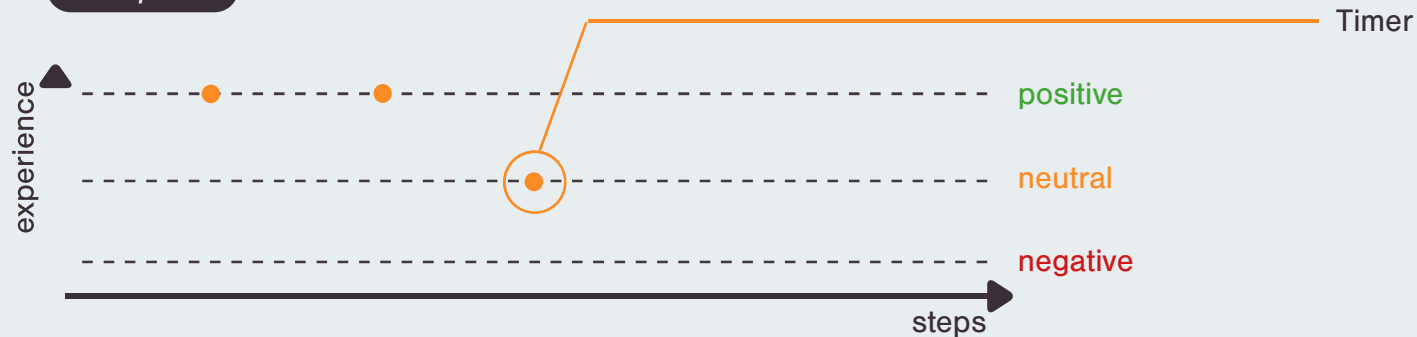
Task 2: primary functions

Participant 1

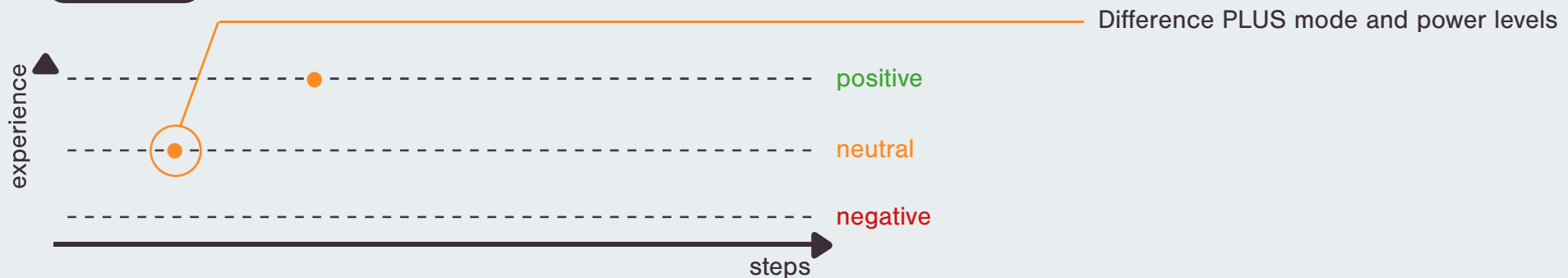
Main pain points



Participant 2



Participant 3



Appendix I: Overall experience graphs

Task 3: PLUS functions

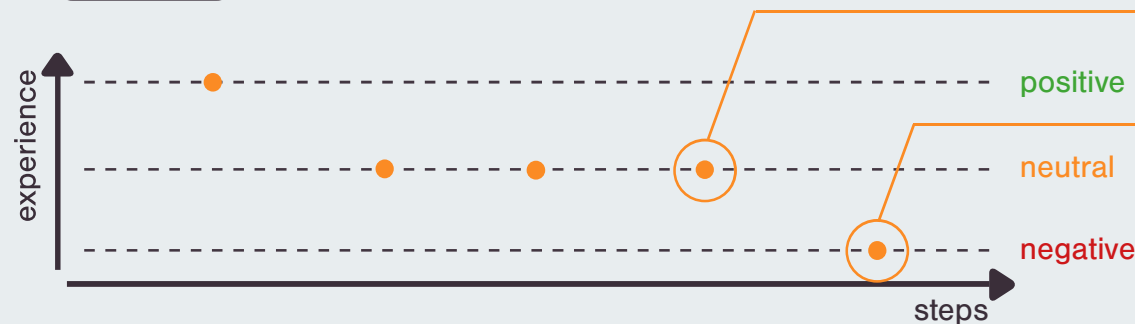
Participant 1



Main pain points

Countdown timer
Frying zone is too hot
Not able to adjust temperatures
Leaving the plus zone
Going to power levels when adjusting temperatures
Not feeling in control

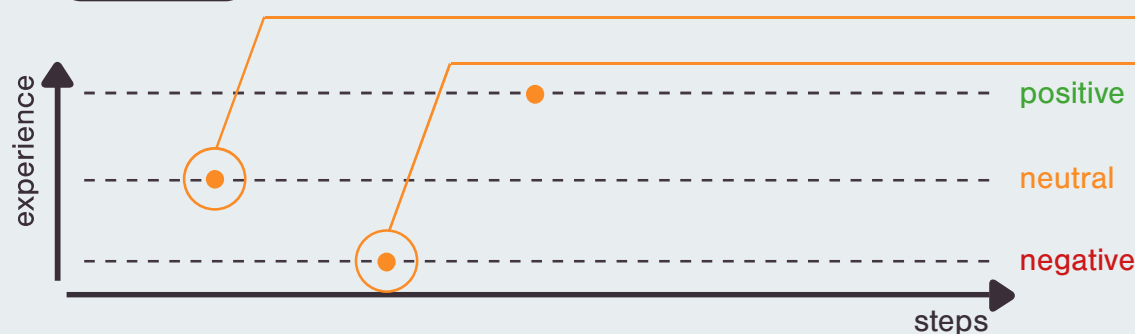
Participant 2



Countdown timer

Frying zone is too hot
Not able to adjust temperatures
Going to power levels when adjusting temperatures
Not feeling in control

Participant 3



Countdown timer

Frying zone is too hot
Not able to adjust temperatures
Going to power levels when adjusting temperatures

Appendix I: Overall experience graphs

Task 4: Celsius functions

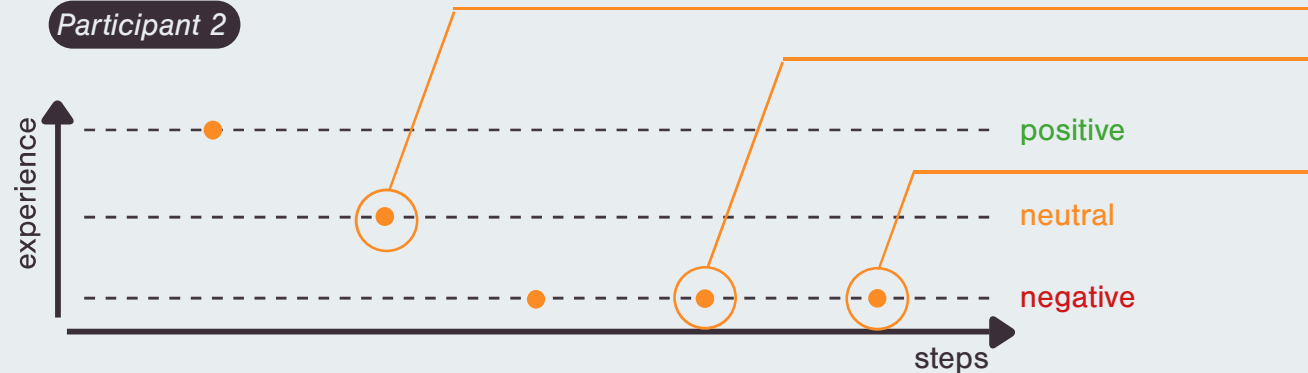
Participant 1

Main pain points



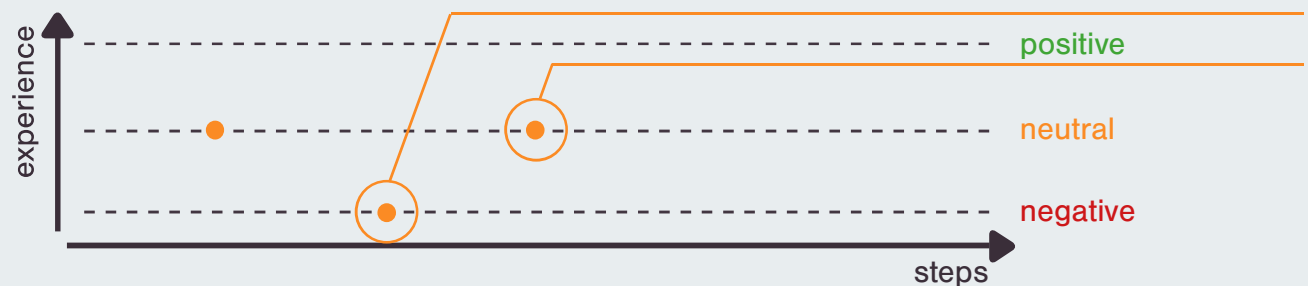
- Connection problems (probe)
- Sound with errors is overwhelming
- Placement of probe
- Confusion about the two temperatures
- Different modes and how to get into them

Participant 2



- Indications on hob unclear (L)
- Connection problems (probe)
- Fluctuation of temperature

Participant 3

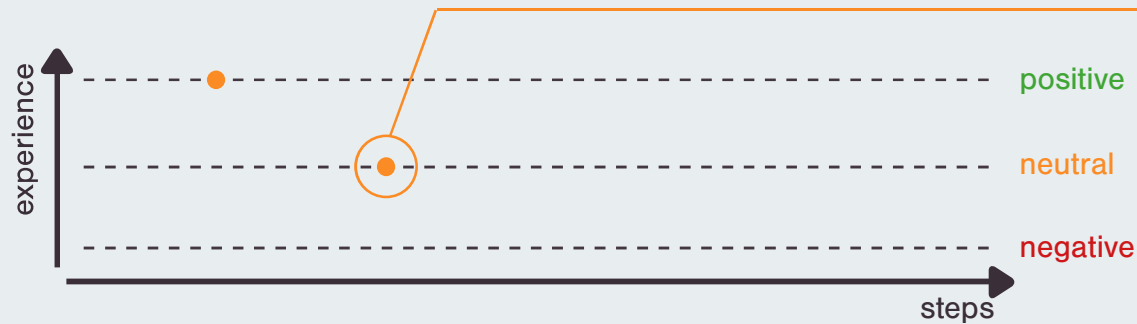


- Connection problems (pan)
- Connection problems (probe)
- Placement of probe
- Sound with errors is overwhelming
- Leaving Celsius mode
- Program pauses whenever probe is touched
- Sliding to desired temperature takes long

Appendix I: Overall experience graphs

Task 5: concluding tasks

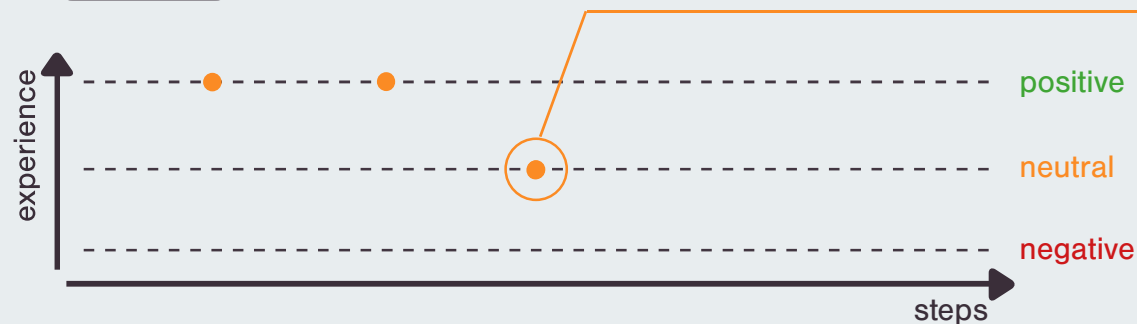
Participant 1



Main pain points

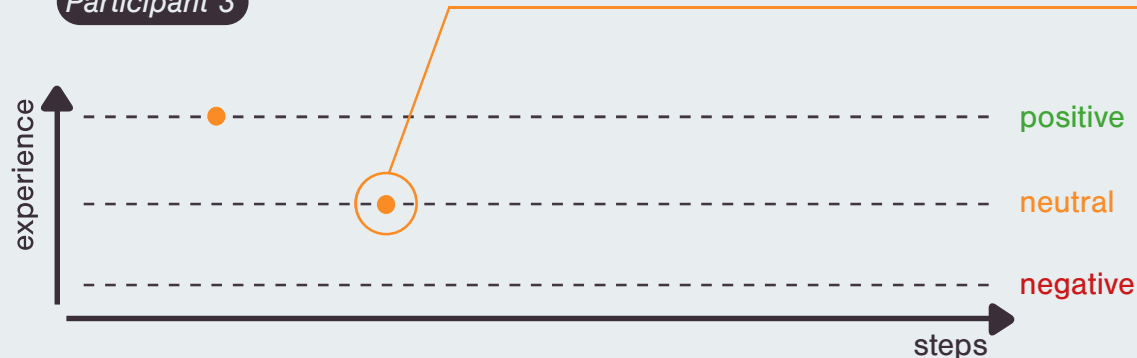
Participants did not know about lock function. Cleaned without using lock function.

Participant 2



Participants did not know about lock function. Cleaned without using lock function. Manual about personal settings was long and unclear

Participant 3



Manual about personal settings was long and unclear
Unable to change settings with pan on hob

Appendix I: Overall experience graphs

Overall pain points

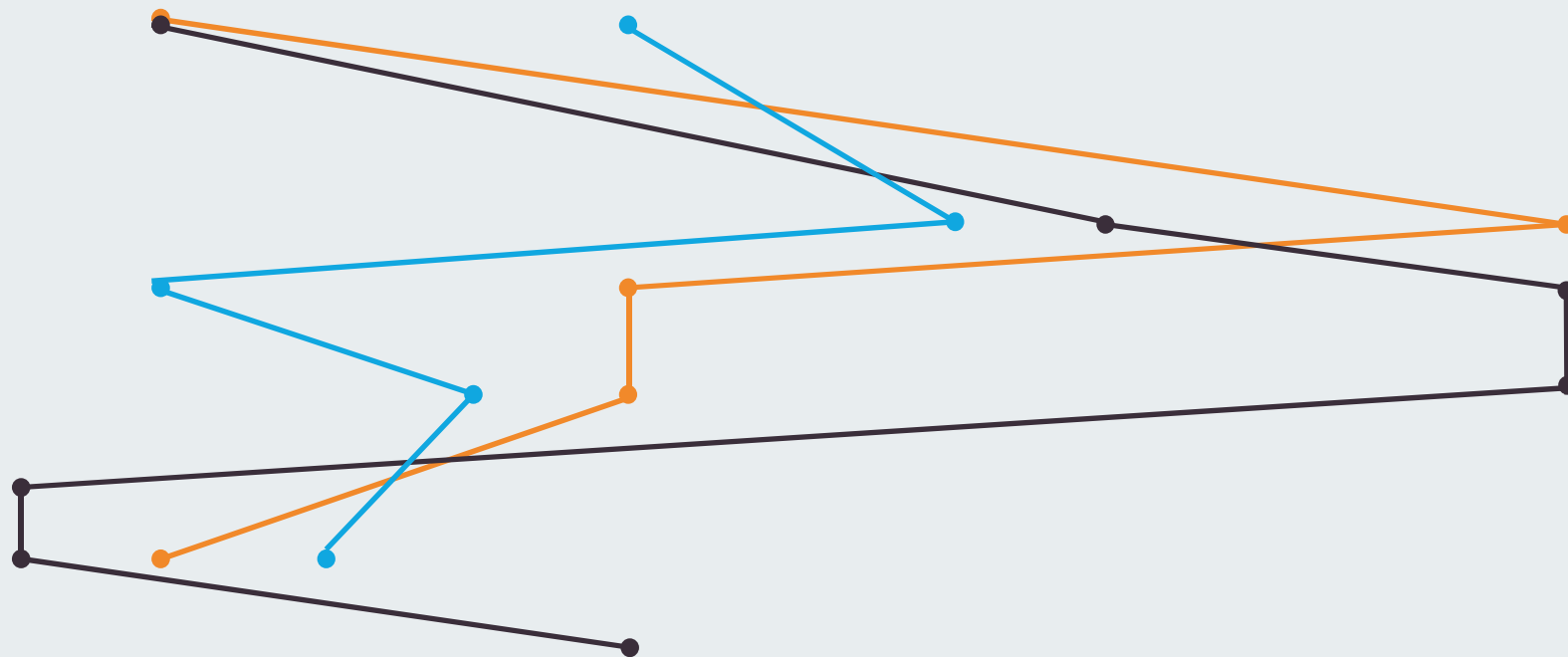
Lock function was unclear and according to participants not necessary (2x)
Pan detection had a short delay
Interface of timer (minutes or hours)
Participants are unclear on the difference between power levels and PLUS mode
Preheat countdown timer (3x)
Frying zone is too hot (3x)
Not able to adjust temperatures (3x)
Leaving the plus zone
Going to power levels when adjusting temperatures (3x)
Not feeling in control (2x)
Connection problems (probe) (3x)
Connection problems (pan)
Placement of probe (2x)
Sound with errors is overwhelming (2x)
Leaving Celsius mode
Program pauses whenever probe is touched
Sliding to desired temperature takes long
Confusion about the two temperatures
Different modes (Celsius, PLUS and power levels) and how to get into them
Fluctuation of temperatures
Participants did not know about lock function. Cleaned without using lock function (2x)
Manual about personal settings was long and unclear (2x)
Unable to change settings with pan on hob

Appendix J: Overall emotion graph

Participant 1

Participant 2

Participant 3



emotions

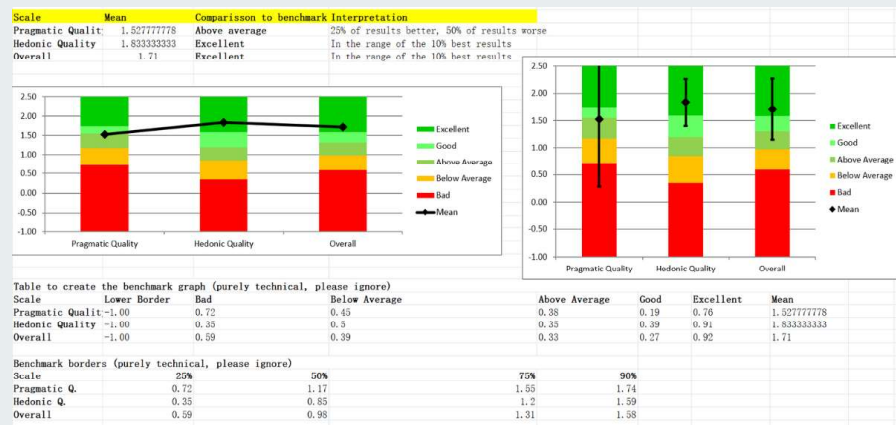


Appendix K: UEQ Questionnaire

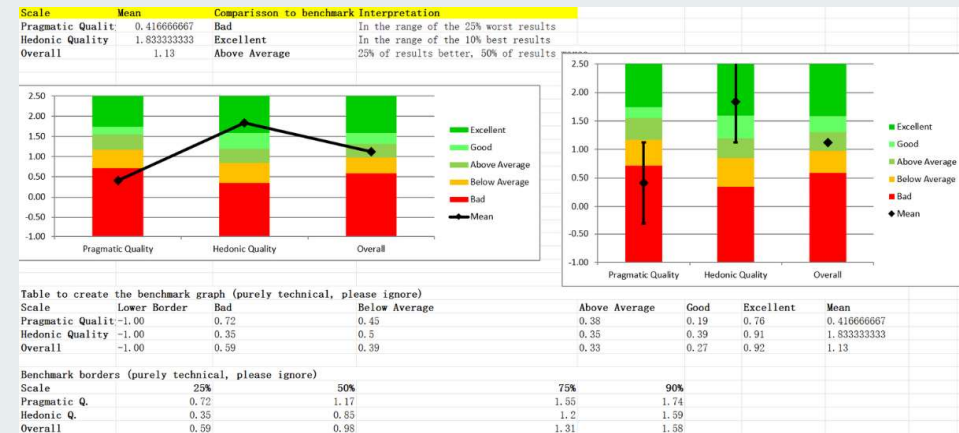
Benchmark

For user testing of the ATAG Celsius Induction Hob, benchmark data, especially scale scores and their confidence intervals, serve two primary purposes. Firstly, they provide a performance evaluation by quantifying the hob's performance in key metrics such as heating speed, energy efficiency, and ease of use, compared to industry standards or competitors, identifying strengths and areas for improvement. Secondly, these data offer an in-depth analysis of user experience, reflecting user satisfaction, convenience of use, and practicality of features. Along with confidence intervals, this further validates the reliability of test results, offering accurate guidance for product design and continuous improvement. Two versions of the benchmark chart are shown. The first one shows only the scale scores in relation to the benchmark categories, the second one shows in addition the confidence intervals of the scale scores.

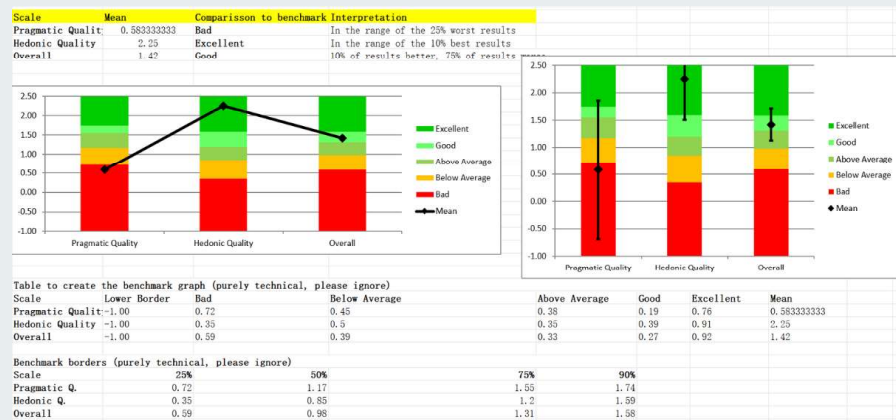
Step 1



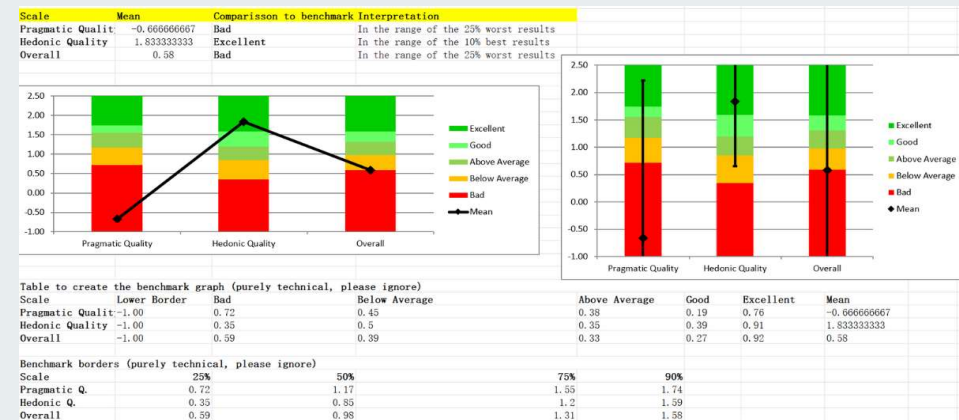
Step 3



Step 2

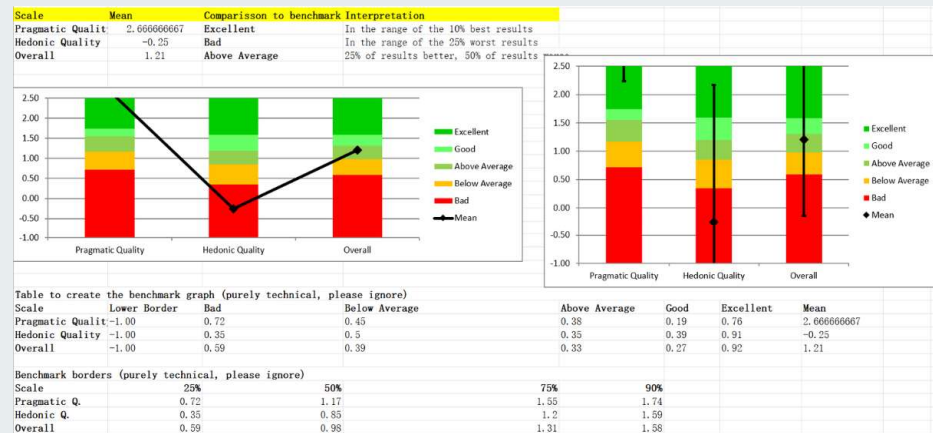


Step 4

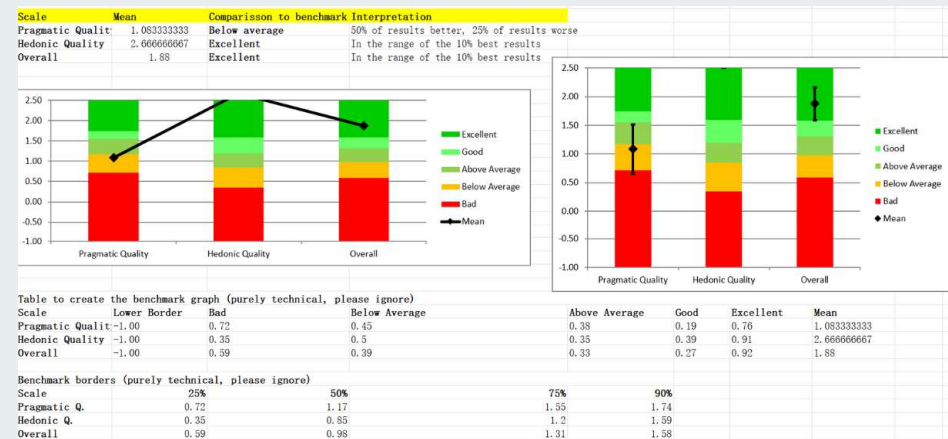


Appendix K: UEQ Questionnaire

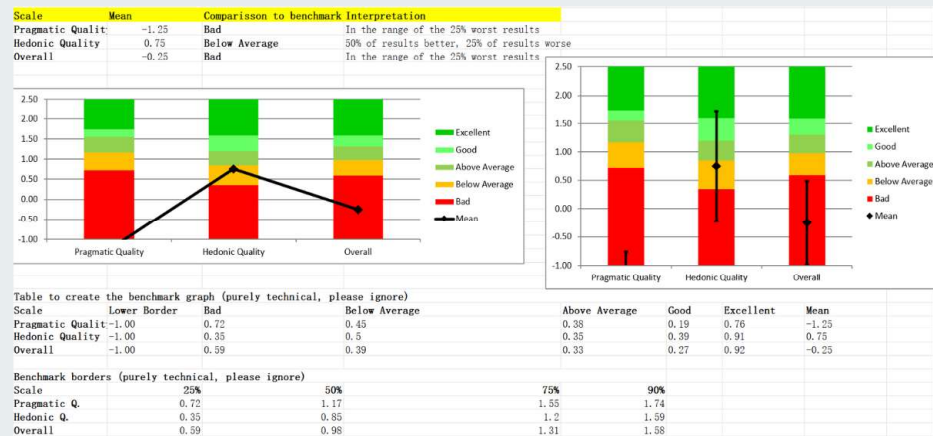
Step 5 - cleaning



Overall



Step 5 - settings

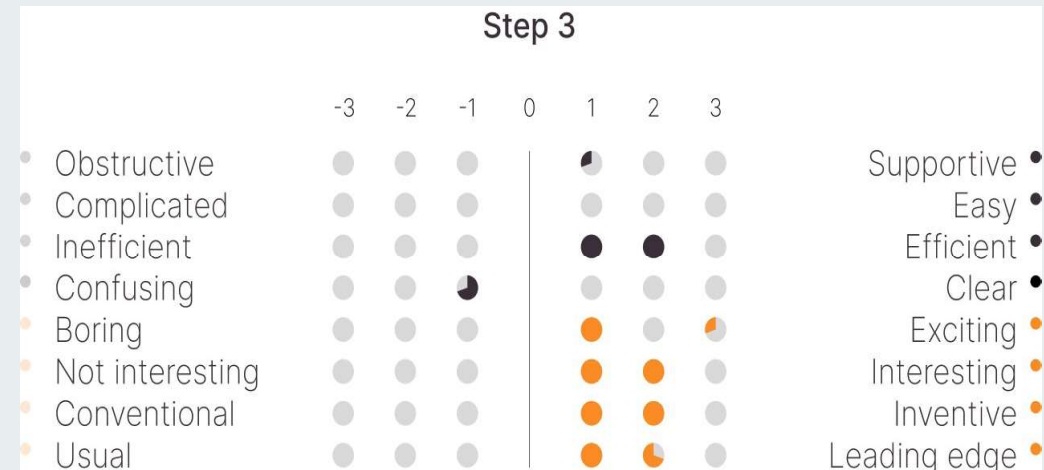
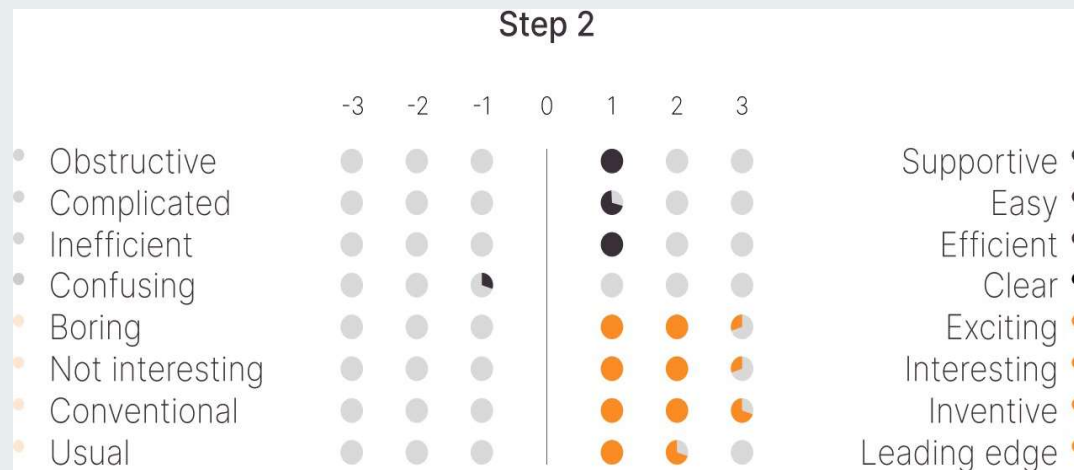
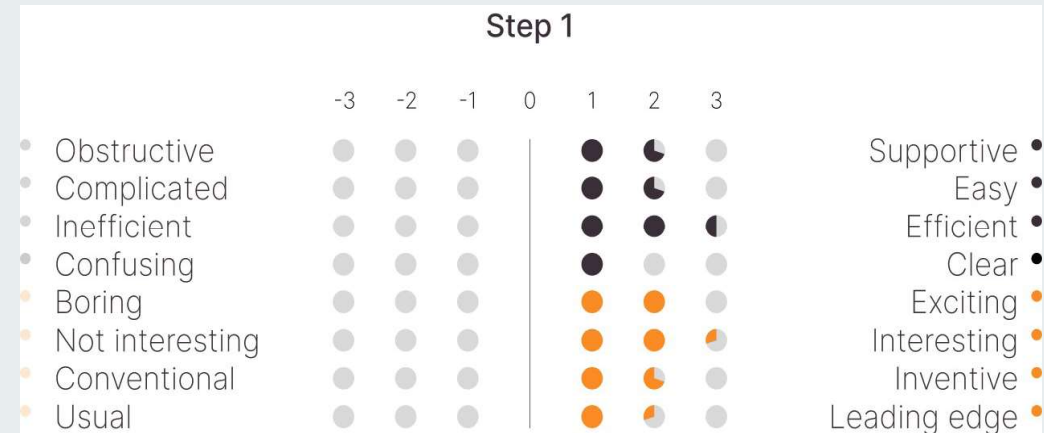


Appendix L: Visuals UEQ Questionnaire

These graphs show an overview of the average values for the UEQ questionnaire for every step. As can be seen from the visualisations in the user test analysis section the ATAG Celsius induction cooker scores above average.

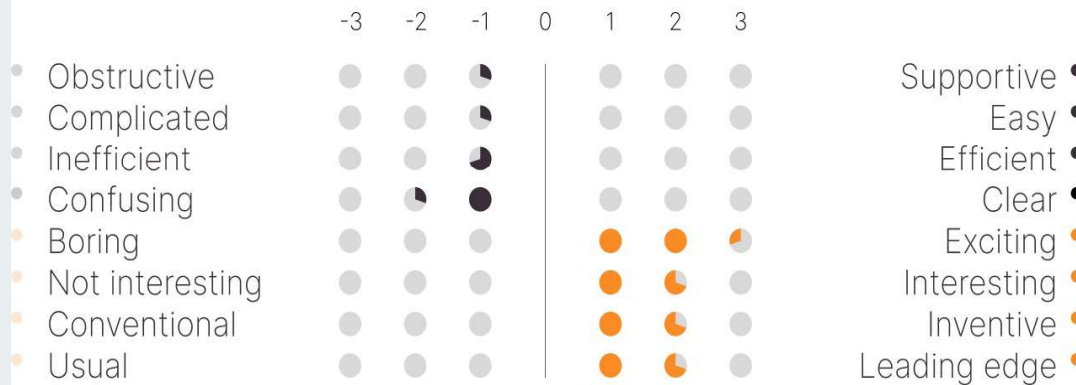
If we look specifically at the pragmatic quality, we can see that this is predominantly due to the cleaning aspects and use of the power levels. The plus cooking and celsius cooking, what makes this product unique compared to other induction hobs, are rated rather low and score below average.

The users score the product high on the hedonic qualities due to its new functions. They however do feel that the product is slightly confusing and complicated.

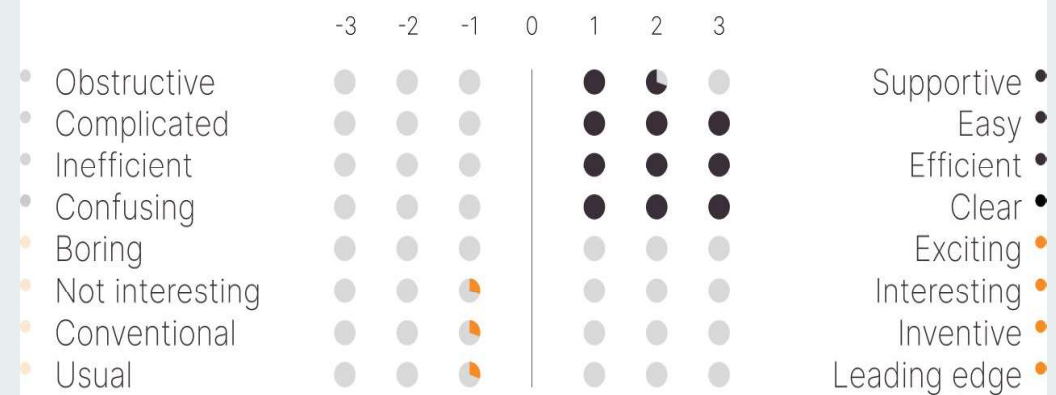


Appendix L: Visuals UEQ Questionnaire

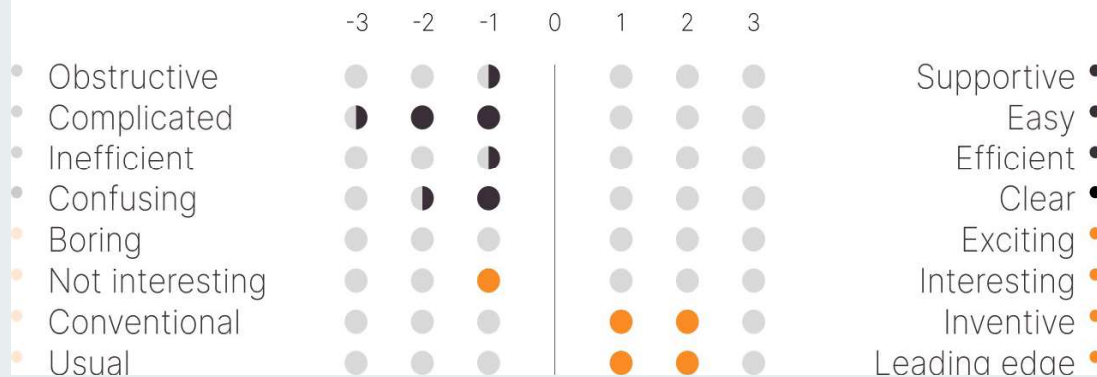
Step 4



Step 5 Cleaning



Step 5 Setting



Overall

